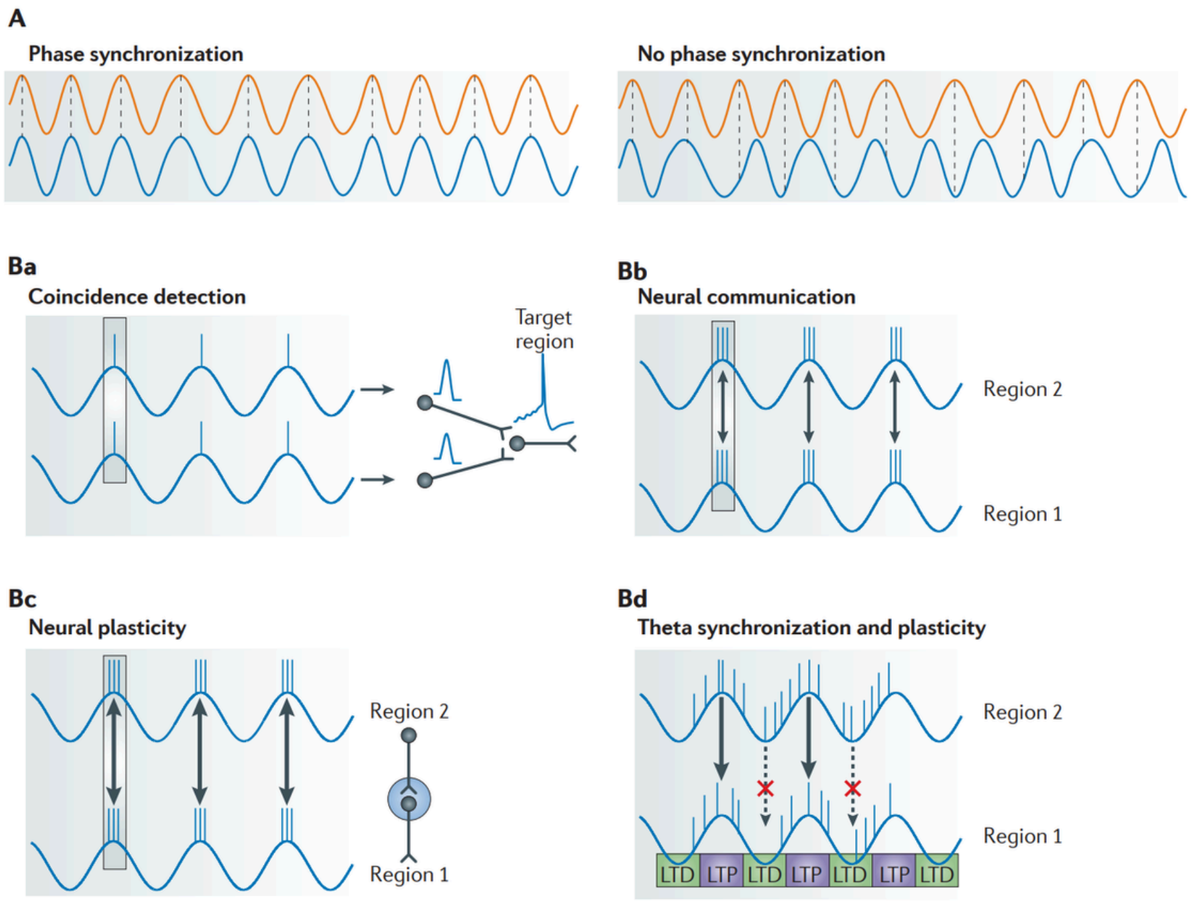
EXPLORING LLS: GRANULARITY, EARLY EXIT AND SCALING LLS

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Start

WHAT IS LLS?

Local Learning Rule inspired by neural synchronization activity that happens in our brain.



Source: The role of phase synchronization in memory processes

HOW LLS?

Each layer projects its own output onto a fixed, untrained basis generated from a periodic function.

Traditional BP:

O(Ln) memory , O(Ln*n) computation

LLS:

O(max(n)) memory , O(Ln*C) computation

$$\nabla_{\mathbf{w}^{(l)}}\mathcal{L} = \frac{\partial \mathcal{L}}{\partial \mathbf{h}^{(l)}}\frac{\partial \mathbf{h}^{(l)}}{\partial \mathbf{z}^{(l)}}\frac{\partial \mathbf{z}^{(l)}}{\partial \mathbf{w}^{(l)}}$$

$$= \frac{\partial \mathcal{L}}{\partial \mathbf{h}^{(L)}}\prod_{i=l+1}^{L}\frac{\partial \mathbf{h}^{(i)}}{\partial \mathbf{h}^{(i-1)}}\frac{\partial \mathbf{h}^{(l)}}{\partial \mathbf{z}^{(l)}}\frac{\partial \mathbf{z}^{(l)}}{\partial \mathbf{w}^{(l)}}$$

1. $\mathbf{h}^{(l)} = f(\mathbf{z}^{(l)})$
2. $\mathbf{h}^{(l)}\mathbf{b}^{(l)\top}$ (Projection on the fixed basis)
3. $\mathbf{p}^{(l)} = \text{softmax}(\mathbf{h}^{(l)}\mathbf{b}^{(l)\top})$
   $\mathcal{L}_{\text{CE}}^{(l)} = \mathbf{y}^{*}\log(\mathbf{p}^{(l)})$
4. $\mathbf{e}^{(l)} = \mathbf{p}^{(l)} - \mathbf{y}^{*}$
5. $\mathbf{e}^{(l)}\mathbf{b}^{(l)}$ (Learning signal for layer l)

$$\Delta\mathbf{w}^{(l)} = \left(\mathbf{e}^{(l)}\mathbf{b}^{(l)} \odot f'(\mathbf{z}^{(l)})\right)^{\top}\mathbf{h}^{(l-1)}$$

$$\mathbf{w}^{(l)} := \mathbf{w}^{(l)} - \eta\Delta\mathbf{w}^{(l)} \quad \text{(Weight update)}$$

Source: Local Learning Rule for Deep Neural Networks Inspired by Neural Activity Synchronization

OUR ATTEMPTS

Per-layer LLS trails BP. Can we close the gap, and what else can the heads do?

Granularity

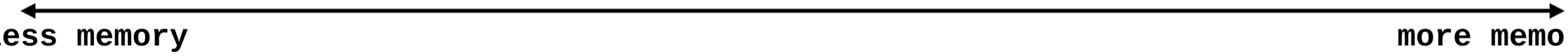
Early exit

Scalability

Attention

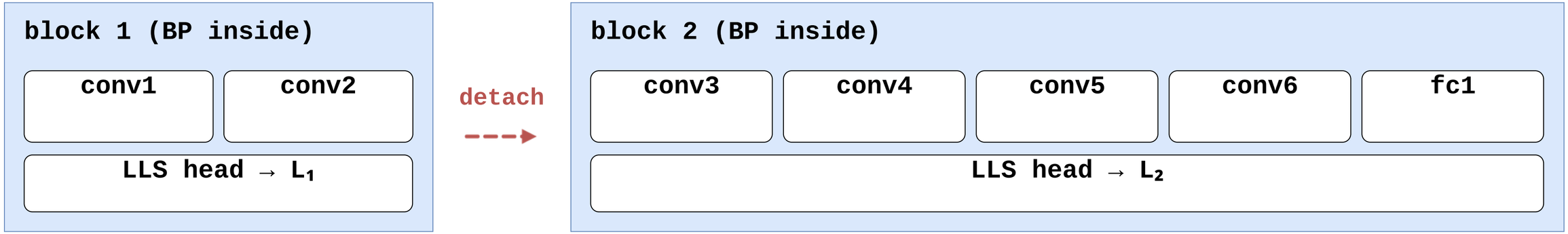
GRANULARITY

1×n	Blocks	(n)
one head per layer: LLS	BP inside, LLS between	one head: BP-like



- How do accuracy and memory change with the number of heads?
- Which cuts are best for a memory budget?
- Does a simple rule transfer between models?

SETUP: PARTITIONS



- (2,5): the first 2 layers, then the other 5.
- (1,1,1,1,1,1,1) is per-layer LLS.
- (7) is one head on a BP network.

SETUP: PARTITIONING SCHEMES

Scheme	Cuts
Uniform	every k units
Pooling-aligned	at resolution changes
Early / late merging	first or last few layers
Exhaustive sweep	all 64 of VGG8
Short-run search	few epochs, keep the best

SETUP: BLOCK-WISE TRAINING

```
def step(self, x: Tensor, y: Tensor) -> Out:
    h = x
    for block, opt in zip(self.blocks, self.opts):
        h = block(h.detach())
        loss, s = block.head(h, y)
        opt.zero_grad(set_to_none=True)
        loss.backward()
        opt.step()
    logits = self.cls(h.detach())
    ...
```

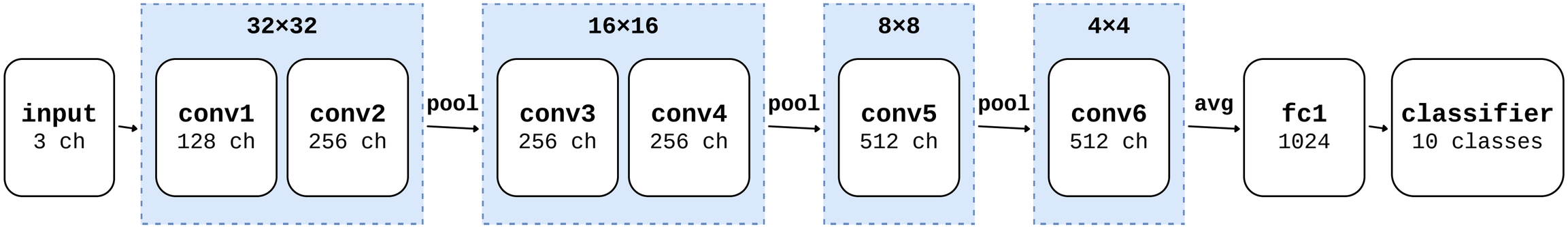
- **Separate LLSHead, with upstream's formula, to attach to any block.**
- **With one layer per block, it matches upstream LLS over training steps.**

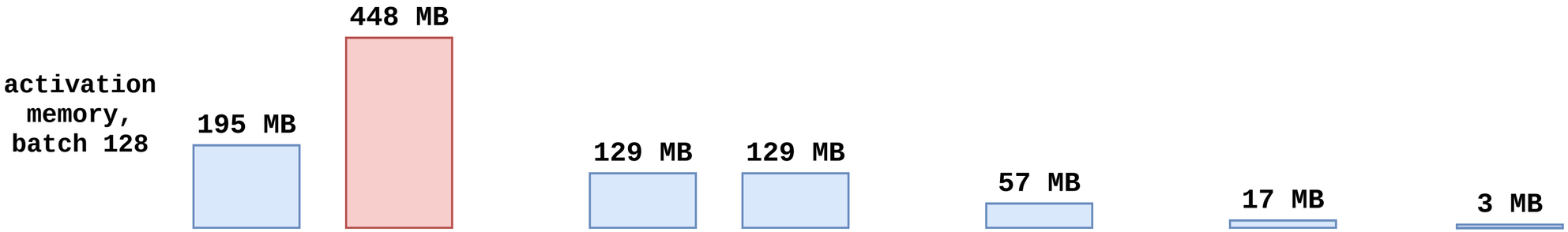
SETUP: TRAINING MEMORY

$$M_{\text{partition}} = \max_{k=1,\ldots,K} M_k, \qquad M_{\text{BP}} = \sum_{\ell=1}^{n} M_\ell$$

```
def saved_bytes(fn: Callable[[], Tensor], static: Iterable[Tensor]) -> int:
    skip = {t.untyped_storage().data_ptr() for t in static}
    seen: dict[int, int] = {}
    def pack(t: Tensor) -> Tensor:
        s = t.untyped_storage()
        if s.data_ptr() not in skip:
            seen[s.data_ptr()] = s.nbytes()
        return t
    with torch.autograd.graph.saved_tensors_hooks(pack, lambda t: t):
        out = fn()
    return sum(seen.values())
```

VGG8: ARCHITECTURE





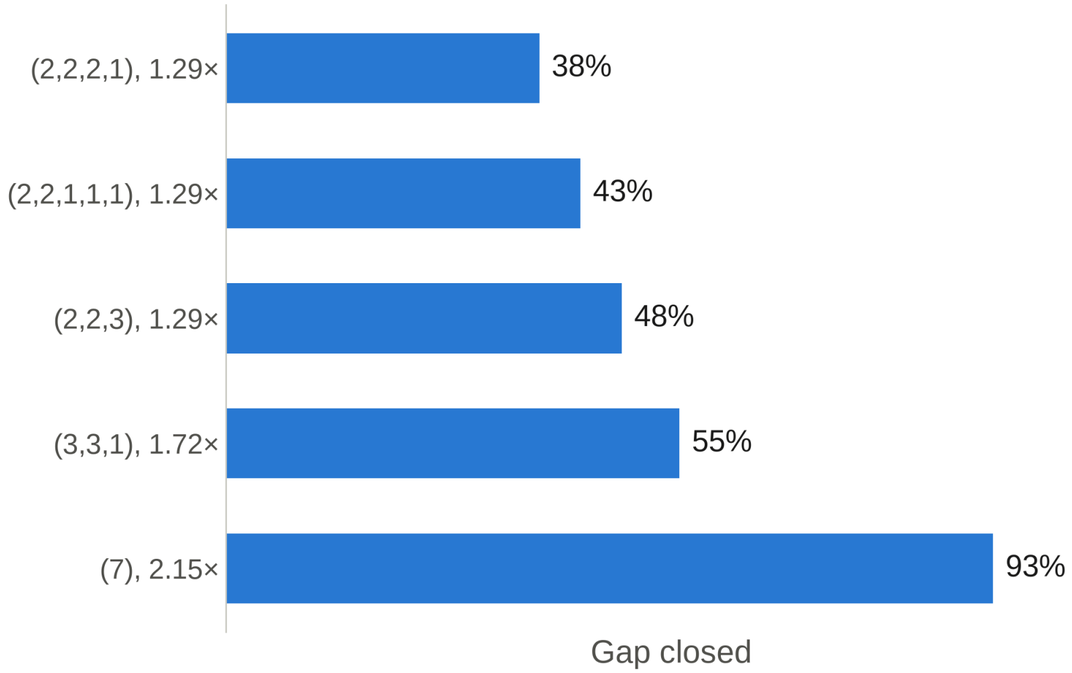
- Peak memory is the largest block, and conv2's block sets it.
- BP keeps every layer's activations at once: 964 MB.

VGG8: MAIN SPLITS

Strategy	Split	Test acc. (%)	Gap closed	Memory (MB)	× LLS
Per-layer LLS	(1,1,1,1,1,1,1)	86.62	0%	448.5	1.00
Uniform k=2	(2,2,2,1)	89.24	38%	578.1	1.29
Pooling-aligned	(2,2,1,1,1)	89.55	43%	578.1	1.29
Random-split	(2,2,3)	89.94	48%	578.1	1.29
Uniform k=3	(3,3,1)	90.41	55%	770.1	1.72
One block	(7)	92.98	93%	964.9	2.15
BP	none	93.48	100%	964.4	2.15

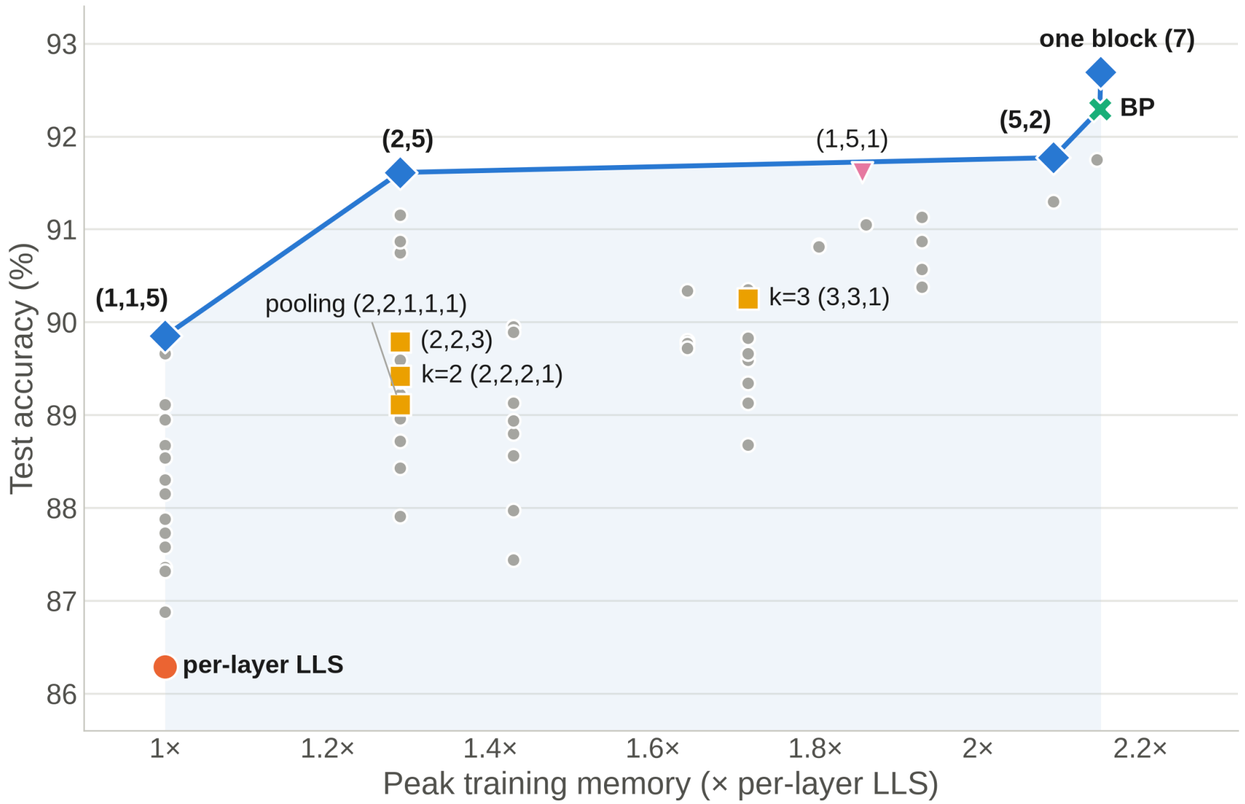
VGG8, CIFAR10, 300 epochs, seed 0. Gap closed is measured between LLS (86.62) and BP (93.48).

VGG8: FIXED STRATEGY



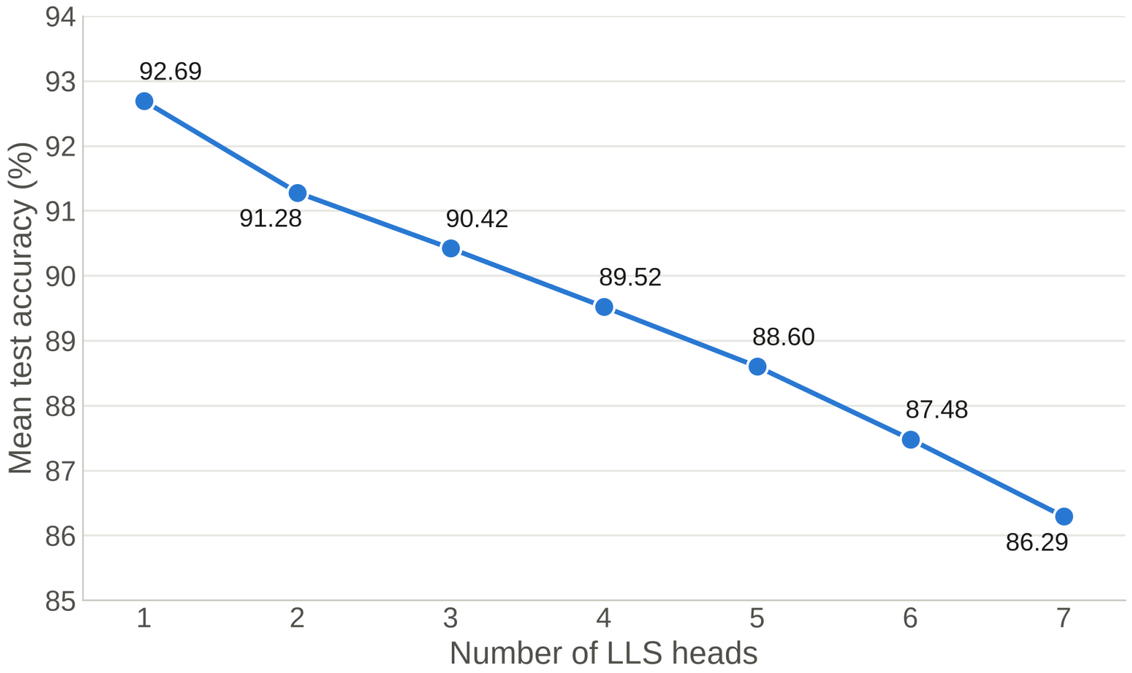
- Every coarser split beats per-layer LLS.
- The (2,2,…) splits all use 578 MB: the largest block sets the peak.
- Fixed rules stop near half the gap; one block reaches 93%.
- One block ≈ BP, so the loss comes from many local heads, not the fixed basis.
- So we ran the full sweep.

VGG8: FULL SWEEP



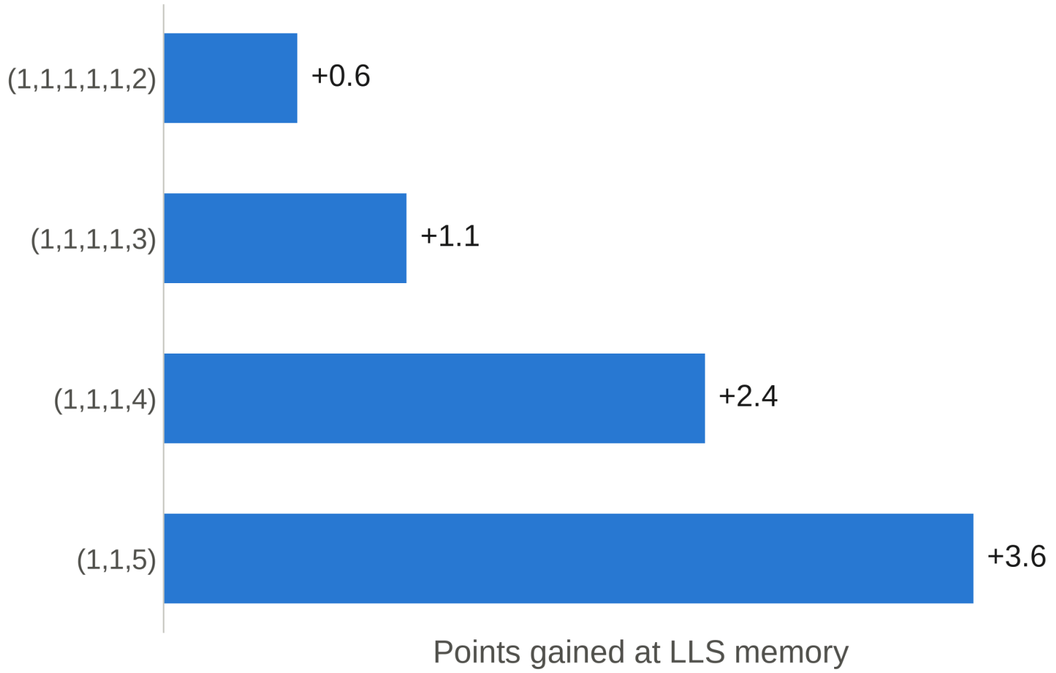
- Trained all 64 partitions.
- 10 columns: memory depends only on conv2's block.
- In one column, accuracy varies by up to 3.7% accuracy.
- Uniform and pooling cuts: 2.2–2.5% accuracy short.

VGG8: FULL SWEEP



- **Rank correlation with accuracy: −0.89.**
- **Each head removed adds about 1% accuracy.**
- **The best split in a column has the fewest heads: (1,1,5), (2,5).**

VGG8: FULL SWEEP



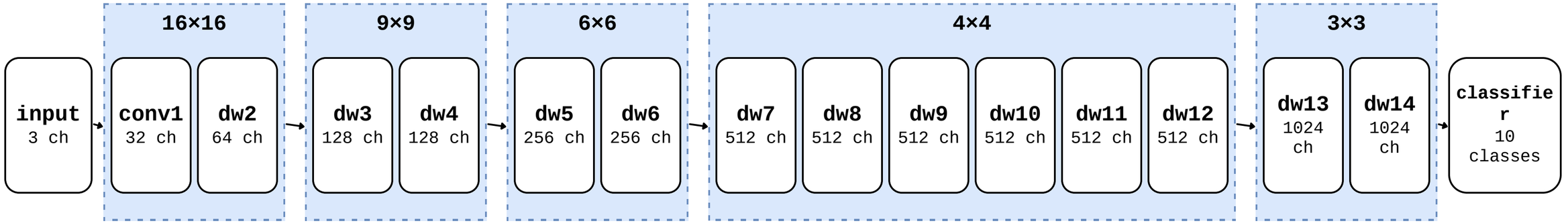
- The last 2–3 layers alone give little.
- The gain grows as the block reaches back to the middle.
- (1,1,5) merges 5 of 7 layers at LLS memory.
- (2,…) adds 1.8 more at the top of its column.

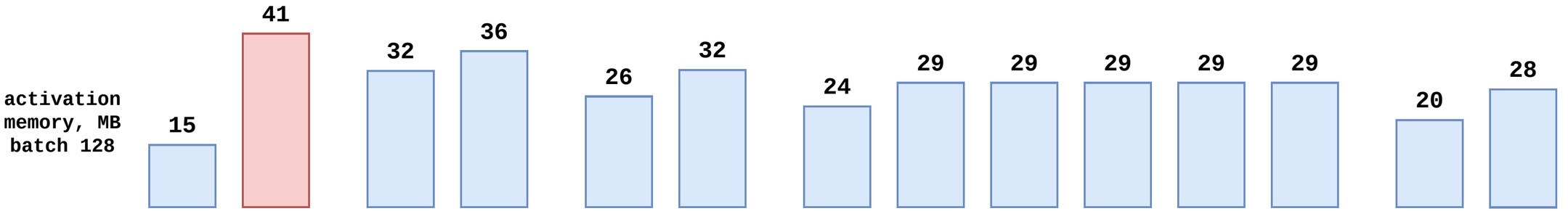
VGG8: FULL SWEEP

Partition	Test acc. (%)	Gap closed	Memory (MB)	× LLS
LLS (1,1,1,1,1,1,1)	86.20	0%	448.5	1.00
(1,1,5)	89.85	60%	448.5	1.00
(2,5)	91.61	89%	578.1	1.29
(5,2)	91.77	91%	938.7	2.09
BP	92.29	100%	964.4	2.15

- **(1,1,5): free gain, small late layers don't raise the peak.**
- **(2,5): the knee.**
- **BP costs 2.15× memory for the last 0.7% accuracy.**
- **(1,5,1) matches (2,5) but needs 1.86×: pair conv2 with conv1.**

MOBILENETV1: ARCHITECTURE





- 14 units: conv1 + 13 depthwise-separable blocks, so 8,192 possible splits.
- Memory is nearly flat, 15–41 MB per layer; BP keeps all of it: 332 MB.

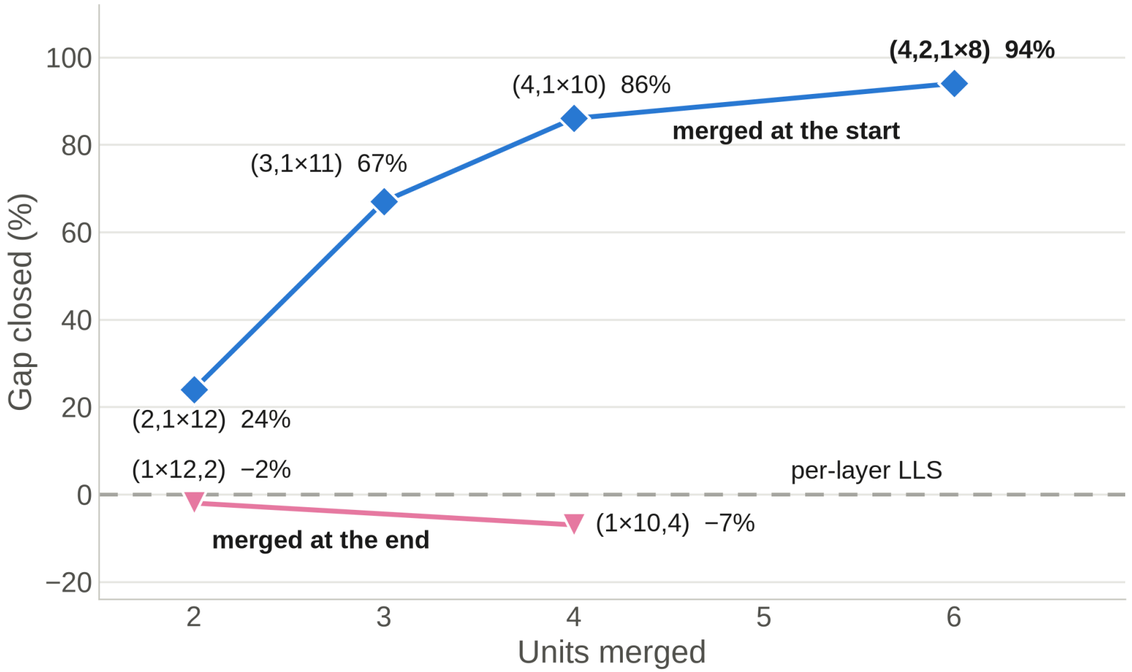
MOBILENETV1: FIXED STRATEGY

Strategy	Split	Heads	Test acc. (%)	Gap closed	Memory (MB)	× LLS
Per-layer LLS	1×14	14	67.82	0%	40.5	1.00
Uniform k=2	2×7	7	77.28	58%	62.2	1.54
Pooling-aligned	(2,2,2,6,2)	5	77.39	59%	144.2	3.56
One block	(14)	1	82.74	92%	332.5	8.21
BP	none	none	84.02	100%	332.5	8.21

MobileNetV1, CIFAR10, 300 epochs, seed 0. Gap closed is measured between LLS (67.82) and BP (84.02).

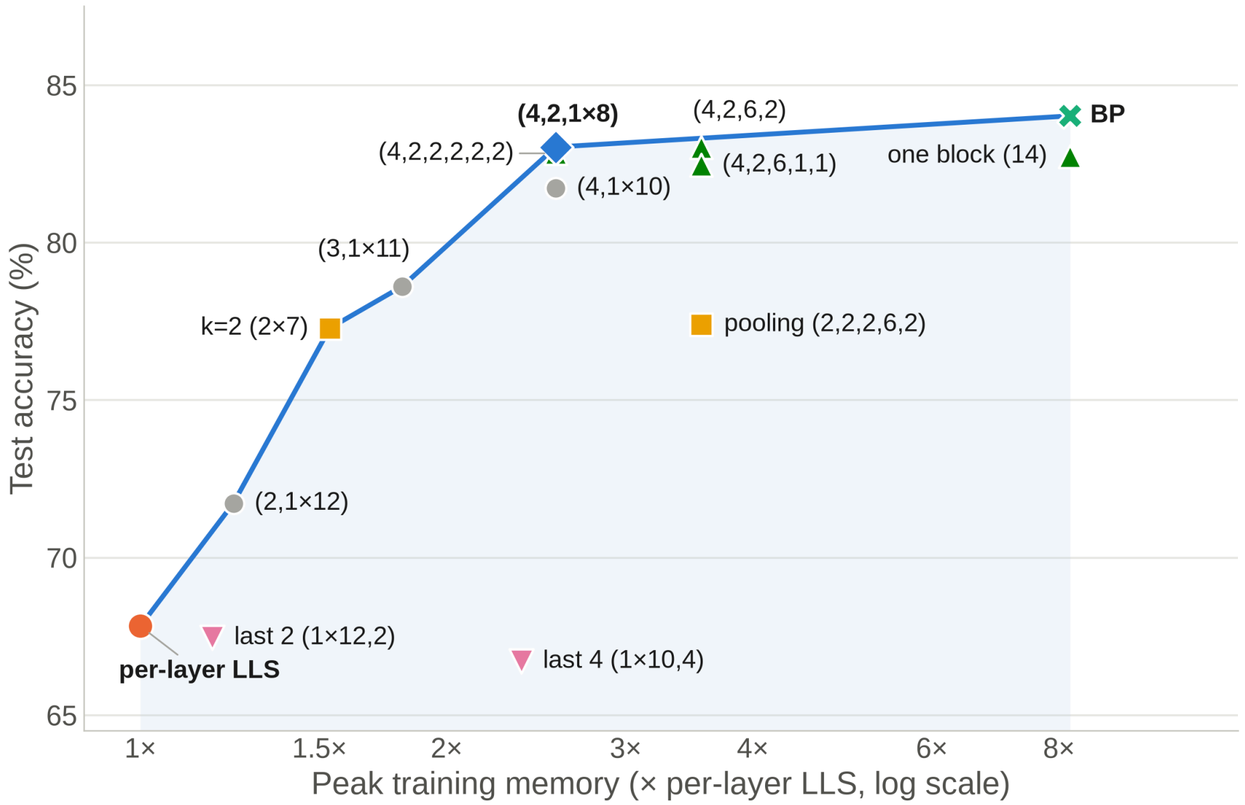
- **The gap is 16.2% accuracy, more than twice VGG8’s 6.9%.**
- **Pooling-aligned matches k=2 at 2.3× its memory: the 6-unit block.**
- **Fixed rules stall at 58–59% again; one block reaches 92%.**

MOBILENETV1: TARGETED SPLITS



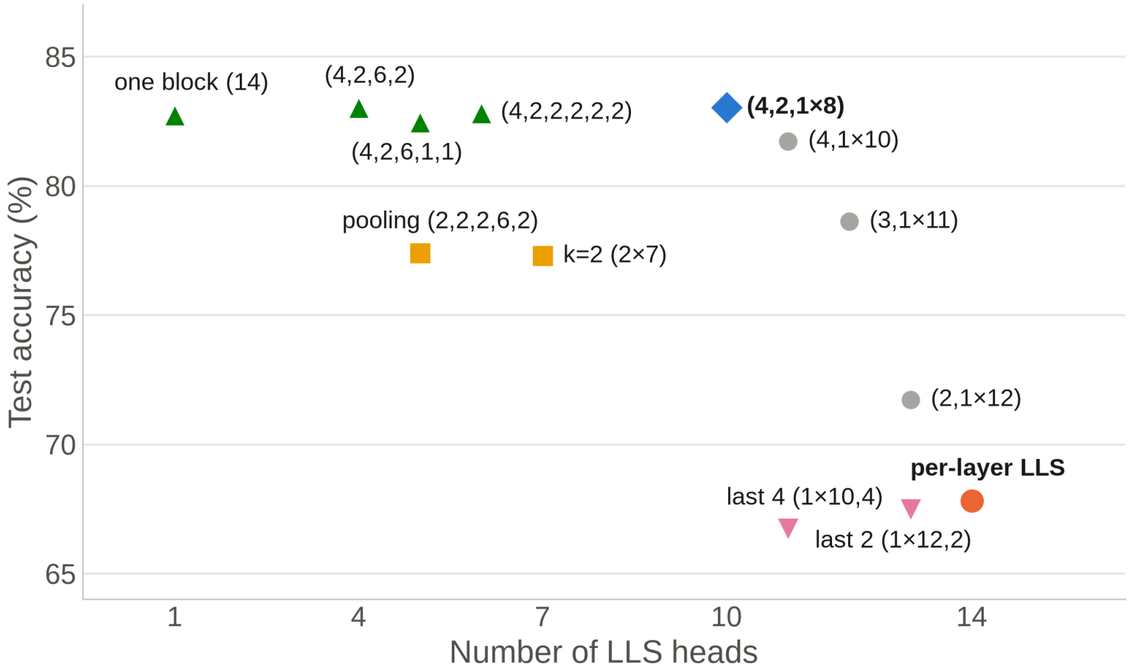
- Merging the last 2 or 4 units: no gain.
- Growing the first block: 24% to 86%.
- (4,2,1×8) reaches 94% at 2.56× LLS.
- The first 6 units decide almost everything.

MOBILENETV1: TARGETED SPLITS



- After (4,2,1×8), nothing improves.
- Further merges stay at 82.5–83.0.
- One block: 0.3 lower at 3.2× the memory.
- BP: 3.2× the memory for the last 1% accuracy.

MOBILENETV1: TARGETED SPLITS



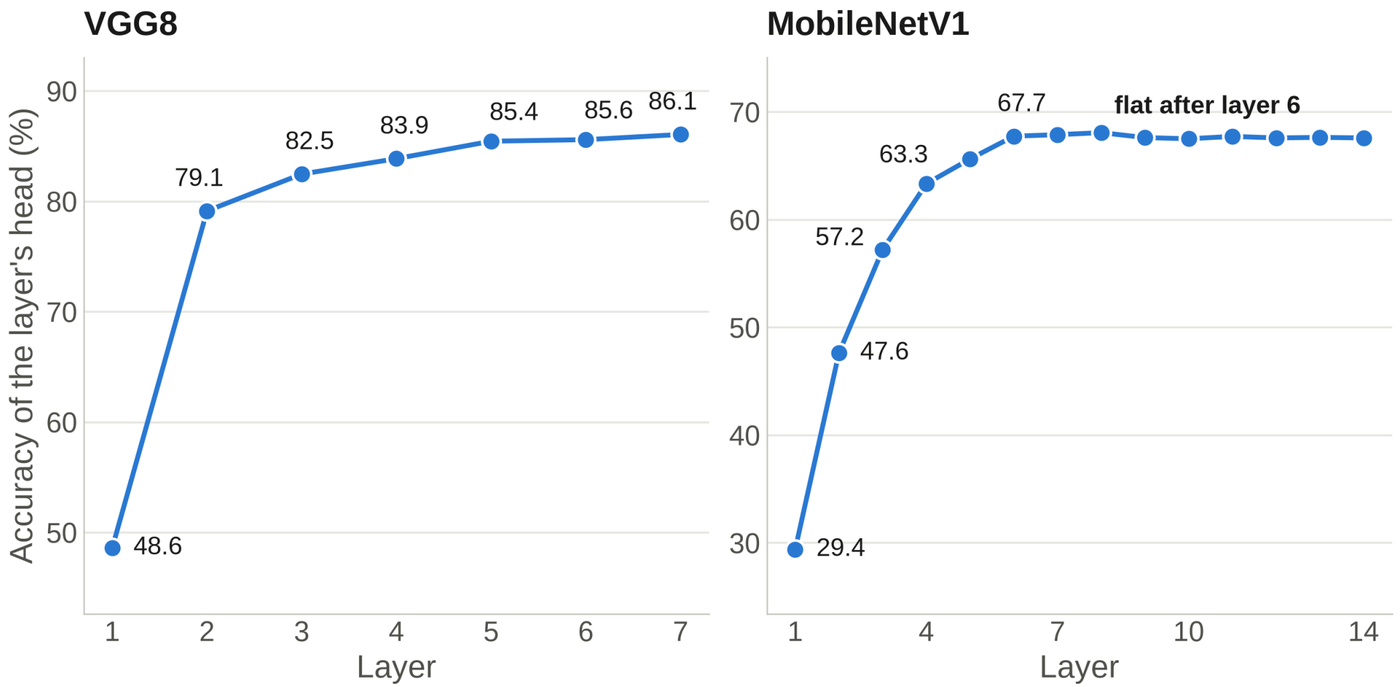
- VGG8: fewer heads, higher accuracy.
- MobileNetV1: where the heads are matters more.
- 10 heads, (4,2,1×8): 83.03.
- 5 heads, pooling-aligned: 77.39.
- 11 heads: 66.7 or 81.7.

MOBILENETV1: TARGETED SPLITS

Partition	Test acc. (%)	Gap closed	Memory (MB)	× LLS
LLS (1×14)	67.82	0%	40.5	1.00
(4,2,1×8)	83.03	94%	103.7	2.56
(4,2,2,2,2,2)	82.83	93%	103.7	2.56
(4,2,6,2)	83.02	94%	144.2	3.56
(4,2,6,1,1)	82.46	90%	144.2	3.56
One block (14)	82.74	92%	332.5	8.21
BP	84.02	100%	332.5	8.21

- **Best: (4,2,1×8) merges only the first 6 units; the other 8 stay per-layer.**
- **One seed: differences under 0.5% accuracy are ties.**
- **Best of the 13 splits we tried, not of all 8,192.**

PER-LAYER ANALYSIS



- Gains come early, then flatten.
- VGG8 – most of the memory is in the first two blocks.
- MobileNet First 6 layers: 45% of BP’s memory.

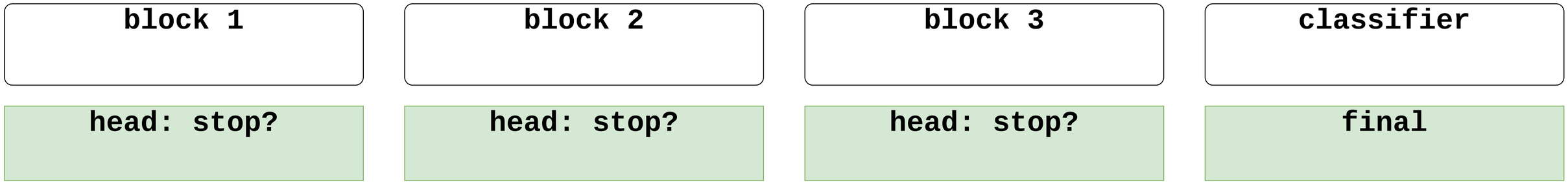
CONCLUSION

89%
of the gap at 1.29×: VGG8

94%
of the gap at 2.56×: MobileNetV1

- Where the cuts go matters more than how many.
- Merging only the last few layers isn't enough on either model.
- VGG8 gains from removing heads; MobileNetV1 from merging its first 6 units.
- On CIFAR100, (2,5) matches BP.
- Simple rules fall 2–6% accuracy short, so search per model.

EARLY EXIT WITH LLS HEADS



- Every block already ends in a classifier, so exits are free.
- K blocks give K+1 exits, and one exit costs one projection.
- We reused the checkpoints and evaluated all rules offline.

SETUP: STOPPING RULES

notation $p_l^{(1)} \geq p_l^{(2)}$: the two largest entries of p_l

confidence $s_l = p_l^{(1)}$

entropy $s_l = -H(p_l)$

margin $s_l = p_l^{(1)} - p_l^{(2)}$

patience $s_l =$ number of consecutive exits ending at l with the same top class

sync (ours) $$s_l = \frac{2}{w(w-1)} \sum_{l-w<i<j\leq l} \cos(p_i, p_j)$$

- **A sample stops at the first exit where s reaches τ.**
- **Also tried: sync+margin, ensembles, skipping early exits.**
- **Baseline: static truncation.**

SETUP: SYNC RULE

$$s_l = \frac{2}{w(w-1)} \sum_{l-w<i<j\leq l} \cos(p_i, p_j) \qquad \cos(p_i, p_j) = \frac{p_i^\top p_j}{\|p_i\| \, \|p_j\|}$$

```
def sync(p: Tensor, w: int) -> Tensor:
    u = F.normalize(p, dim=-1)
    g = u @ u.transpose(1, 2)
    out = torch.full(p.shape[:2], float("-inf"))
    for e in range(w - 1, p.shape[1]):
        blk = g[:, e - w + 1:e + 1, e - w + 1:e + 1]
        out[:, e] = (blk.sum((1, 2)) - w) / (w * (w - 1))
    return out
```

- **Intuition: once heads agree, the representation has settled.**
- **One overconfident head can't trigger an exit.**

SETUP: COST AND OPERATING POINT

$$\text{MAC fraction} = \frac{1}{F}\,\mathbb{E}_x\left[B_{l(x)} + \sum_{i \le l(x)} h_i + \text{decisions}\right]$$

Symbol	Meaning
x, Ex	One test sample; E averages over all samples
l(x)	The exit where x stops
Bl(x)	Backbone MACs up to that exit
hi	MACs of head i, paid for every head checked up to l(x)
decisions	Operations for the stop test at each checked exit
F	MACs of the full network, without heads

- **δ = 1% accuracy: cheapest setting on validation, same setting on test.**
- **We compared with linear probes and with BP frozen and joint exits.**

SETUP: MAC FRACTION

VGG8, margin τ = 0.77: MAC fraction 0.60

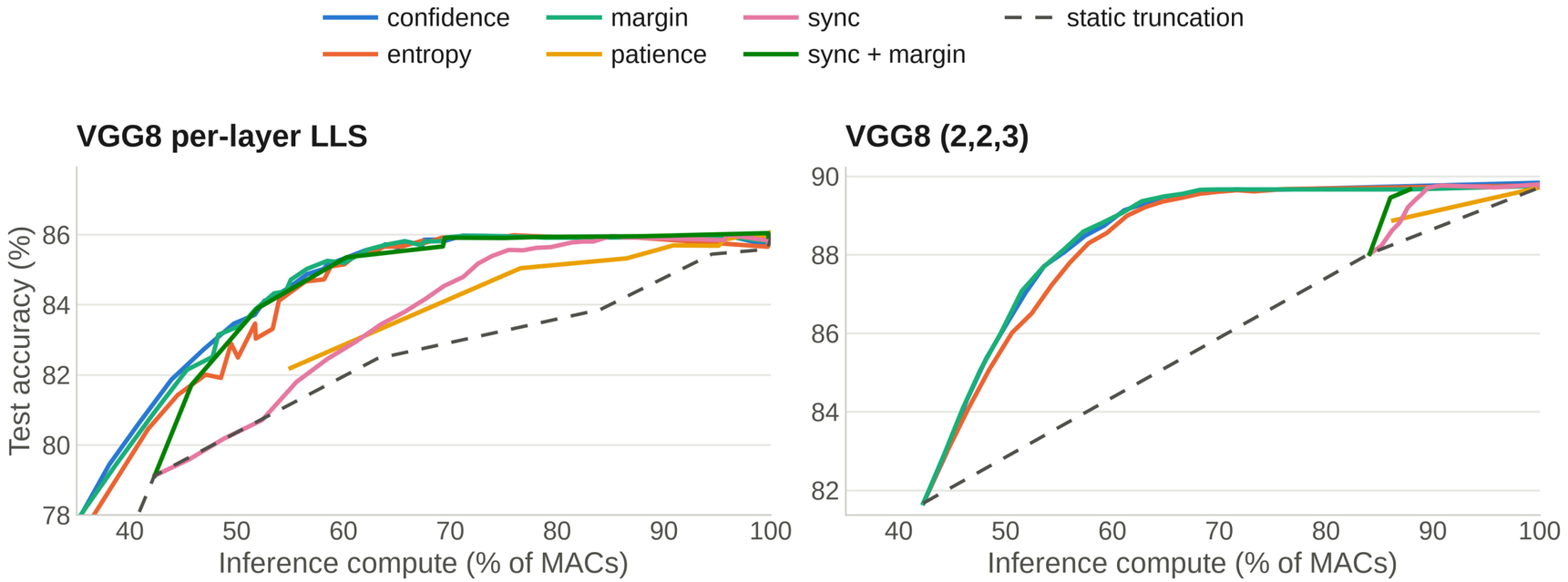
Exit at	Compute	Exiting here
conv1	0.5%	813
conv2	42%	4,568
conv3	63%	1,274
conv4	84%	387
conv5	95%	412
conv6	99.7%	216
fc1	100%	127
classifier (final)	100%	2,203

MobileNetV1, margin τ = 0.12: MAC fraction 0.12

Exit at	Compute	Exiting here
conv1–dw4 skipped	0.6–6.7%	0
dw5	9.4%	8,327
dw6	15%	999
dw7	20%	309
dw8	29%	132
dw9–dw12	39–68%	90
dw13–dw14	79–100%	9
classifier (final)	100%	134

- **Each head: stop if top probability − second probability ≥ τ; else continue.**
- **MAC fraction = mean of “Compute so far” over the 10,000 test images.**
- **VGG8: 46% stop at conv2, 22% never exit. MobileNetV1: 83% stop at dw5.**

STOPPING RULES



- **Heads are already calibrated: ECE 0.02–0.08.**
- **So a single head's margin is usually enough.**

STOPPING RULES: CHOSEN SETTINGS

Rule	VGG8 setting	MACs	MobileNetV1 setting	MACs
Static truncation	stop at exit 5	0.94	stop at exit 6	0.15
Patience	n = 4	0.91	n = 3, from exit 3	0.16
Sync	w = 2, τ = 0.992	0.74	w = 3, τ = 0.931, from exit 2	0.13
Sync + margin	w = 2, τ = 0.857, m = 0.68	0.69	w = 2, τ = 0.875, m = 0.07	0.12
Entropy	τ = 0.56	0.61	τ = 2.28, from exit 6	0.15
Confidence	τ = 0.80, from exit 2	0.60	τ = 0.13, from exit 6	0.15
Margin	τ = 0.77	0.60	τ = 0.12, from exit 5	0.12

Per-layer LLS, δ = 1%, fitted temperature, single-exit probabilities.

- **Margin wins on VGG8; sync does not beat it.**
- **On MobileNetV1, margin and sync + margin tie at 0.12.**
- **Confidence and entropy reduce to static truncation there: τ barely filters.**

STOPPING RULES: THRESHOLDS

SEARCH RANGE (PER-LAYER LLS)

Rule	VGG8	Median	MN	Median
Confidence τ	0.15–1.00	0.91	0.13–1.00	0.63
Entropy τ	0–2.25	0.38	0–2.28	1.05
Margin τ	0–1.00	0.86	0–1.00	0.43
Patience n	2–8	–	2–15	–
Sync τ (w = 2)	0.02–1.00	0.999	0.09–1.00	0.998

MARGIN τ CHOSEN

Split	τ	MACs
VGG8 per-layer	0.77	0.60
VGG8 (2,2,3)	0.69	0.61
VGG8 (3,3,1)	0.36	0.68
MN per-layer	0.12	0.12
MN (2,1×12)	0.18	0.10
MN (4,2,6,2)	0.19	0.10

- **τ is not set by hand: 41 values taken from validation scores.**
- **VGG8 needs a strict margin (0.77); MobileNetV1 a loose one (0.12).**
- **Sync values crowd near 1, so sync is hard to tune.**

COMPUTE SAVINGS: VGG8

Split	Memory (× LLS)	No exits	With exits	Accuracy lost	Compute saved	Best rule
BP (baseline)	2.15	93.31%	–	–	0%	none
Per-layer	1.00	85.92%	85.19%	0.73%	39.9%	margin, τ = 0.77
(2,2,1,1,1)	1.29	88.98%	88.04%	0.94%	42.6%	margin, τ = 0.59
(2,2,2,1)	1.29	89.23%	88.36%	0.87%	42.1%	margin, τ = 0.56 (ens2)
(2,2,3)	1.29	89.71%	88.74%	0.97%	40.8%	confidence, τ = 0.78
(3,3,1)	1.72	90.04%	89.23%	0.81%	32.2%	margin, τ = 0.34 (ens2)

Test set, δ = 1% accuracy, last-epoch checkpoints. ens, ens2: rule applied to averaged exits.

- **Compute saved: MACs skipped at inference, against running the full network (0%).**
- **Accuracy lost = no exits − with exits.**
- **(2,2,…) saves most: 41–43% at 1.29× training memory.**
- **(3,3,1) saves less: its first exit comes only at conv3.**

COMPUTE SAVINGS: MOBILENETV1

Split	Memory (× LLS)	No exits	With exits	Accuracy lost	Compute saved	Best rule
BP (baseline)	8.21	84.24%	–	–	0%	none
Per-layer	1.00	67.64%	67.18%	0.46%	88.1%	sync + margin, τ = 0.87
(1×12,2)	1.18	67.56%	66.25%	1.31%	89.6%	sync + margin, τ = 0.48 (ens2)
(1×10,4)	2.37	67.90%	66.38%	1.52%	89.1%	sync, τ = 0.98 (ens)
(2,1×12)	1.23	72.15%	70.61%	1.54%	90.0%	margin, τ = 0.18
2×7	1.54	77.40%	76.71%	0.69%	85.5%	sync, τ = 0.89
(2,2,2,6,2)	3.56	77.40%	75.89%	1.51%	89.7%	margin, τ = 0.14

Test set, δ = 1% accuracy, last-epoch checkpoints. ens, ens2: rule applied to averaged exits.

- **Every split saves 85–90%, whatever its accuracy: the architecture sets this.**
- **Late merges save the same compute but add no accuracy.**

MOBILENETV1: EARLY MERGES

Split	Memory (× LLS)	No exits	With exits	Accuracy lost	Compute saved	Best rule
BP (baseline)	8.21	84.24%	–	–	0%	none
(3,1×11)	1.81	78.79%	76.91%	1.88%	92.6%	margin, τ = 0.09 (ens2)
(4,1×10)	2.56	82.01%	81.10%	0.91%	92.7%	margin, τ = 0.13
(4,2,1×8)	2.56	83.31%	82.76%	0.55%	88.3%	confidence, τ = 0.55
(4,2,2,2,2,2)	2.56	83.11%	82.15%	0.96%	91.0%	margin, τ = 0.19
(4,2,6,1,1)	3.56	82.52%	82.09%	0.43%	89.9%	margin, τ = 0.19
(4,2,6,2)	3.56	82.81%	82.34%	0.47%	89.9%	margin, τ = 0.19

Test set, δ = 1% accuracy, last-epoch checkpoints. ens, ens2: rule applied to averaged exits.

- **(4,2,1×8) is best: 82.76% with exits, 88.3% saved, at 2.56× memory.**
- **Its best rule is confidence, not margin.**
- **Every early merge saves 88–93%.**

COMPUTE SAVINGS: ACCURACY LOST

VGG8 CIFAR100

Split	No exits	With exits	Accuracy lost	Compute saved
Per-layer	54.55%	53.39%	1.16%	19.9%
(1,1,5)	63.66%	62.28%	1.38%	8.9%
(2,2,1,1,1)	58.80%	57.37%	1.43%	24.3%
(2,5)	67.27%	65.65%	1.62%	15.7%

SPLITS LOSING OVER 1% ON TEST

Set	Lost over 1%	Worst
VGG8 main	0 of 5	0.97%
MobileNetV1	5 of 12	1.88%
VGG8 CIFAR100	4 of 4	1.62%
VGG8, 64 splits	35 of 64	2.20%

- We allow 1% loss when choosing the setting on validation.
- On new test images, the loss can go slightly over: mean 1.01% over 64 splits.
- Compute saved stays steady; only accuracy shifts.

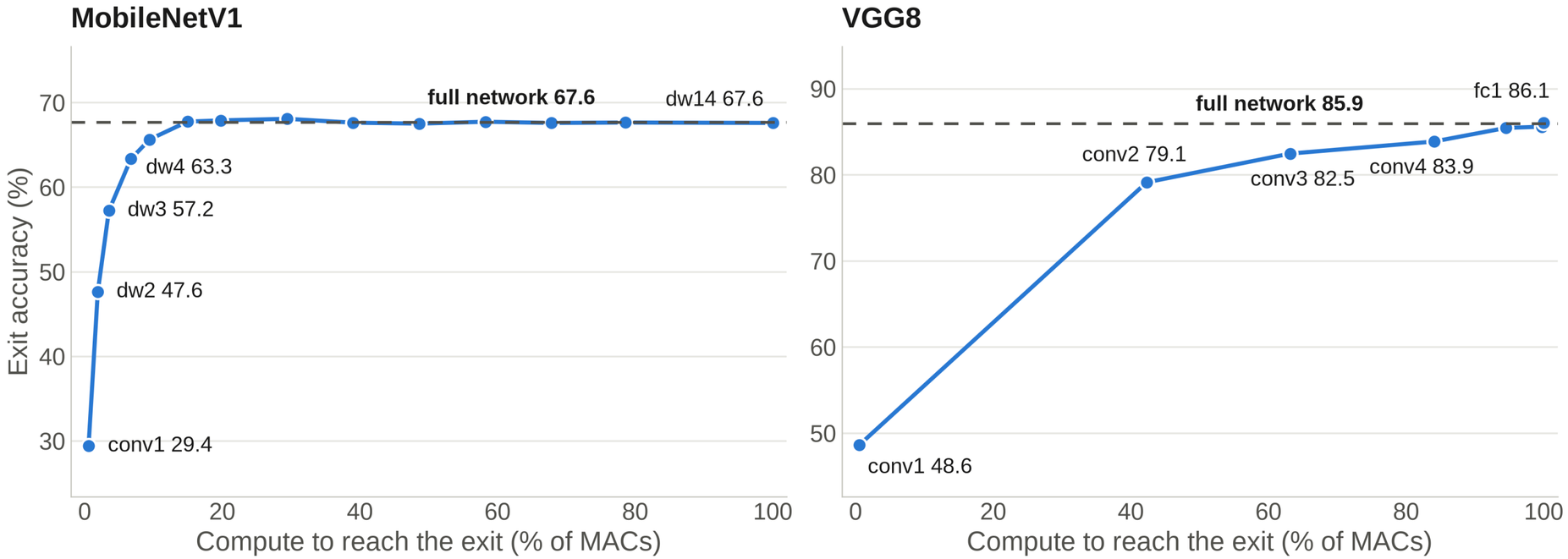
COMPUTE SAVINGS: TRADE-OFF

δ (accuracy %)	VGG8 test acc.	VGG8 MACs	MobileNetV1 acc.	MobileNetV1 MACs
0	86.06	0.74	67.95	0.16
0.5	85.71	0.64	67.56	0.15
1	85.19	0.60	67.18	0.12
2	84.38	0.55	66.19	0.09

Per-layer LLS. Full network on test: VGG8 85.92, MobileNetV1 67.64.

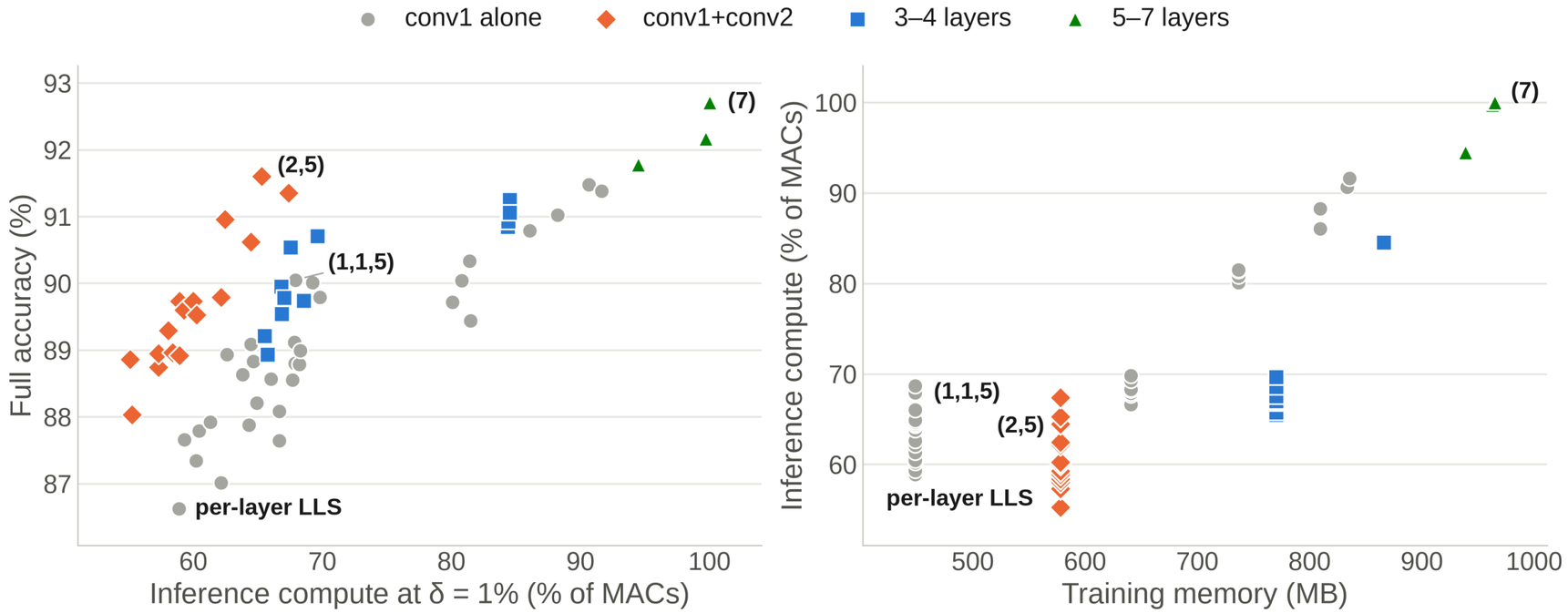
- **Giving up more accuracy buys more compute.**
- **Even with no allowed loss: VGG8 0.74, MobileNetV1 0.16.**
- **Past 1%, each extra 1% of accuracy buys only 3–5% of compute.**

COMPUTE SAVINGS: EXIT PROFILE



- **An exit only saves the compute after it.**
- **So the architecture, not the rule, bounds the savings.**

GRANULARITY AND EXITS



- **Fewer, stronger exits beat many weak ones.**
- **(4,2,6,2): 82.3% at 10% of MACs vs 67.2% at 12%.**

GRANULARITY AND EXITS: BEST SPLITS

	TRAINING: GRANULARITY			INFERENCE: EARLY EXIT			
Split	Memory (× LLS)	No exits	Gap closed	With exits	Accuracy lost	Compute saved	Best rule
VGG8 per-layer	1.00	86.62%	0%	85.15%	1.47%	41.1%	confidence
VGG8 (1,1,5)	1.00	90.05%	60%	88.54%	1.51%	32.1%	margin
VGG8 (2,5)	1.29	91.60%	89%	90.55%	1.05%	34.7%	confidence
VGG8 (5,2)	2.09	91.78%	91%	91.50%	0.28%	5.5%	static
MobileNetV1 per-layer	1.00	67.64%	0%	67.18%	0.46%	88.1%	sync + margin
MobileNetV1 (4,2,1×8)	2.56	83.31%	94%	82.76%	0.55%	88.3%	confidence
MobileNetV1 (4,2,6,2)	3.56	82.81%	94%	82.34%	0.47%	89.9%	margin

Test set. Gap closed: share of the per-layer LLS to BP accuracy gap recovered by merging layers.

- **Training helps accuracy: (2,5) closes 89% of the gap at 1.29× memory.**
- **Exits then help compute: (2,5) saves 35%; (4,2,1×8) saves 88%.**
- **Merging more closes more gap but removes exits: (5,2) saves only 5.5%.**

COMPUTE SAVINGS: SUMMARY

Network	Split	No exits	With exits	Accuracy lost	Compute saved	Speedup	Best rule
MobileNetV1	(4,2,1×8)	83.31%	82.76%	0.55%	88.3%	8.6×	confidence
VGG8	(3,3,1)	90.04%	89.23%	0.81%	32.2%	1.5×	margin
VGG8 CIFAR100	(2,5)	67.27%	65.65%	1.62%	15.7%	1.2×	confidence

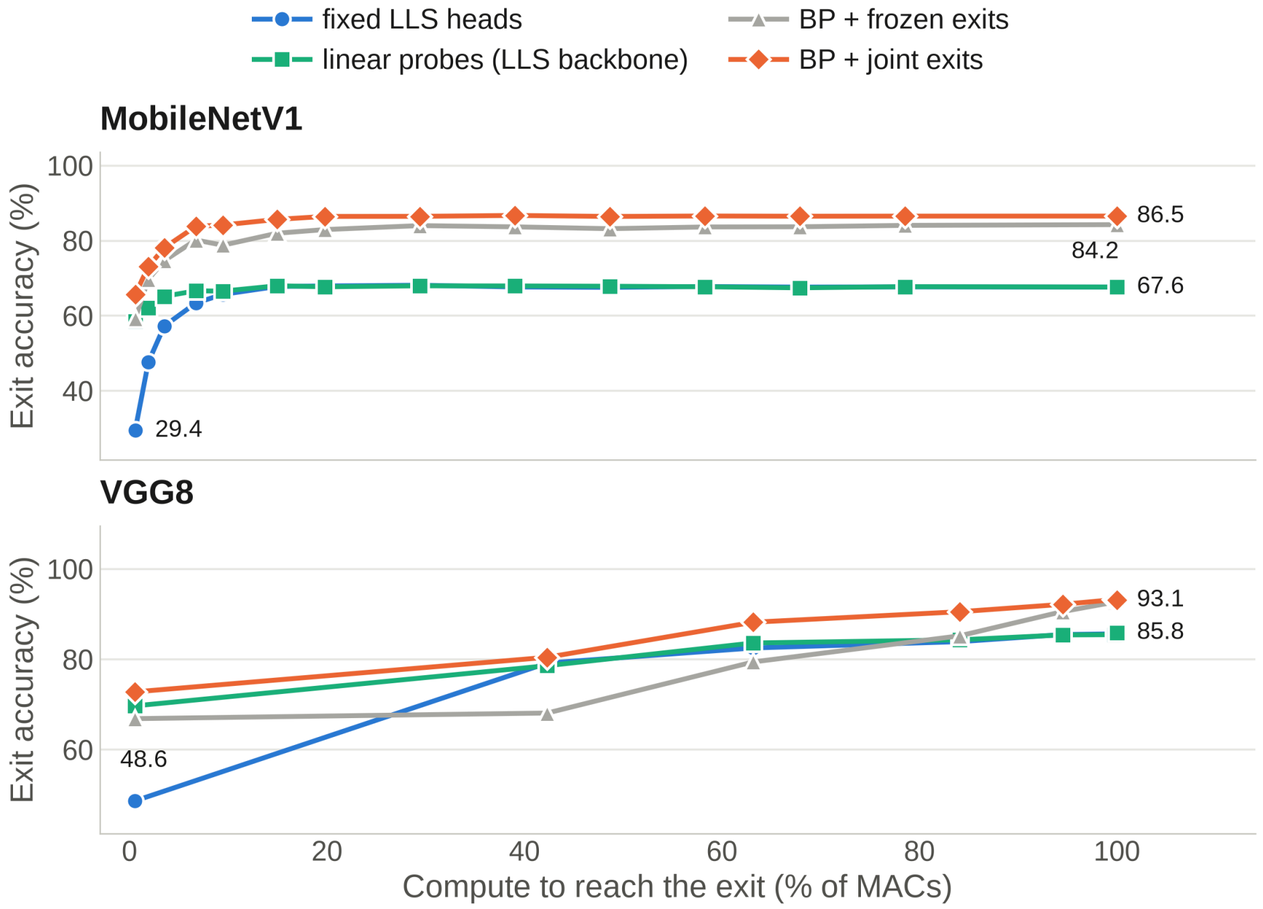
8.6×
MobileNetV1

1.5×
VGG8

1.2×
VGG8 CIFAR100

- **Each row: that network’s most accurate split with early exits.**

FIXED VS LEARNED EXITS



- **Fixed heads nearly match linear probes, with no training.**
- **BP joint exits stay best.**
- **LLS's advantage is training memory.**

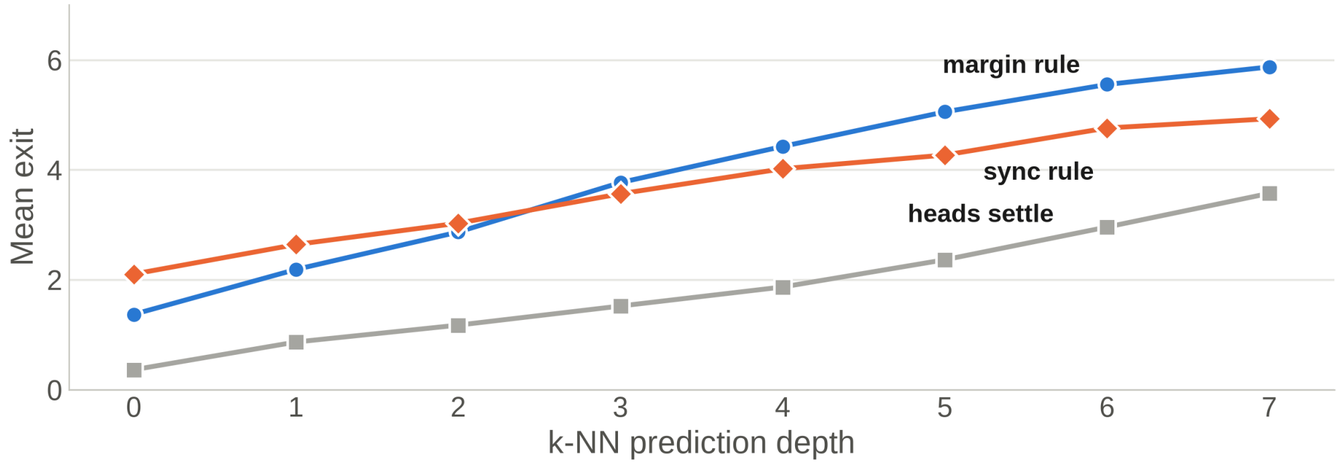
FIXED VS LEARNED EXITS: BASELINES

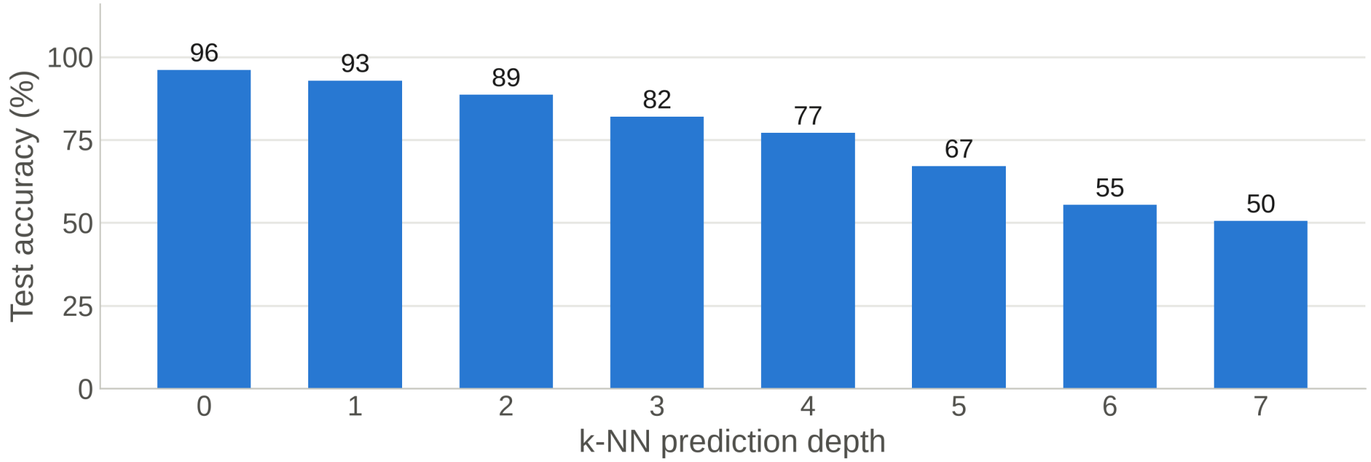
Exit type	Extra training	Memory (× LLS)	No exits	With exits	Accuracy lost	Compute saved
VGG8 fixed (3,3,1)	none	1.72	90.04%	89.23%	0.81%	32.2%
VGG8 probes (3,3,1)	30 epochs	1.72	90.04%	89.30%	0.74%	33.0%
VGG8 BP frozen	30 epochs	2.15	93.31%	92.10%	1.21%	30.1%
VGG8 BP joint	full run, 100 epochs	≥ 2.15	93.05%	91.79%	1.26%	47.9%
MobileNetV1 fixed (4,2,6,2)	none	3.56	82.81%	82.34%	0.47%	89.9%
MobileNetV1 probes (4,2,6,2)	30 epochs	3.56	82.81%	82.20%	0.61%	92.5%
MobileNetV1 BP frozen	30 epochs	8.21	84.24%	82.83%	1.41%	90.2%
MobileNetV1 BP joint	full run, 300 epochs	≥ 8.21	86.50%	85.62%	0.88%	91.1%

Test set, δ = 1% accuracy. Baseline for fixed heads: linear probes on the same backbone.

- **Fixed heads match trained probes within 0.2% accuracy, with no training.**
- **Probes save at most 3% more compute.**
- **BP is more accurate, but needs 1.25–2.3× the training memory.**

EXIT QUALITY AND DIFFICULTY





- The basis is nearly orthogonal: largest |cos| 0.046.
- Weak exits come from cat vs dog: 253 of 256 heads.
- Difficulty: k-NN prediction depth.
- Spearman 0.44–0.59 on VGG8.

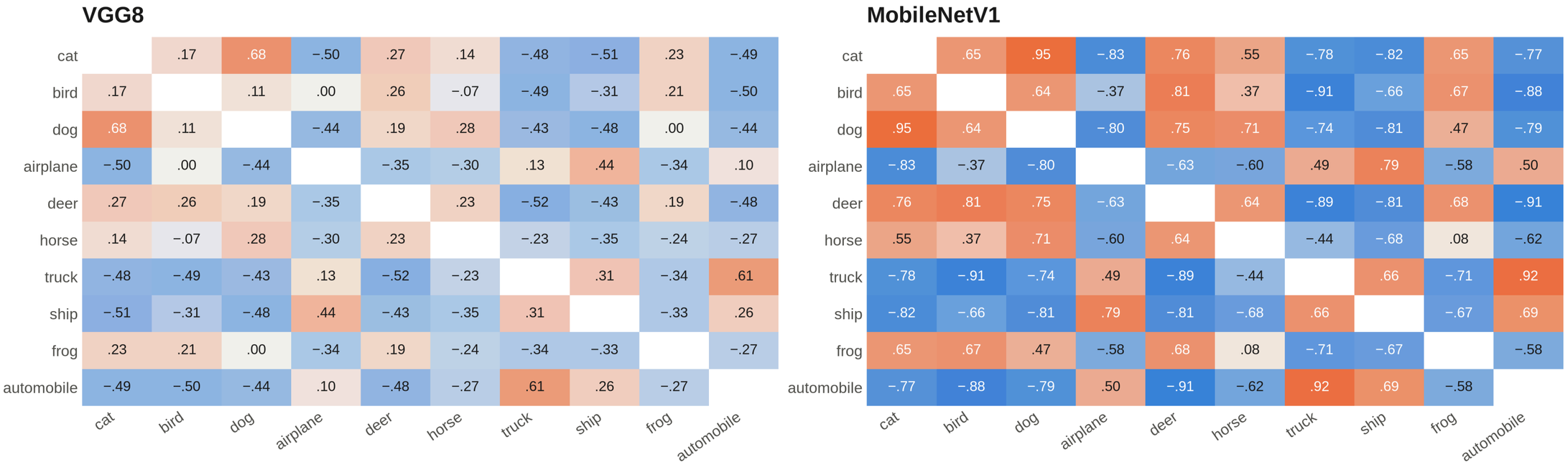
EXIT QUALITY: CLASS DIFFICULTY

Class	VGG8 acc.	VGG8 depth	MN acc.	MN depth	Basis |cos|	Closest class	VGG8 output	MN output
Cat	71.1%	3.14	43.9%	9.08	0.011	dog	0.68	0.95
Bird	79.7%	2.83	54.3%	7.76	0.015	deer	0.26	0.81
Dog	80.6%	2.19	57.9%	8.13	0.041	cat	0.68	0.95
Airplane	87.5%	2.17	74.1%	5.05	0.037	ship	0.44	0.79
Deer	85.7%	2.00	63.4%	7.11	0.010	cat, bird	0.27	0.81
Horse	89.6%	1.73	72.8%	6.02	0.010	dog	0.28	0.71
Truck	89.2%	1.27	72.8%	5.31	0.020	automobile	0.61	0.92
Ship	90.9%	1.20	78.0%	4.14	0.009	airplane	0.44	0.79
Frog	92.1%	1.12	81.0%	4.79	0.014	cat, deer	0.23	0.68
Automobile	92.8%	1.10	78.2%	5.04	0.038	truck	0.61	0.92

Per-layer LLS, test. Depth: low = easy. Basis, output: cosine to the closest class. MN: MobileNetV1.

- **Basis does not explain difficulty: rank correlation 0.06 on VGG8, 0.16 on MobileNetV1.**
- **Cat is hardest, yet its basis row is among the most orthogonal.**
- **Learned output closeness fits better: 0.21 on VGG8, 0.65 on MobileNetV1.**

EXIT QUALITY: CLASS ORTHOGONALITY



Per-layer LLS, last head, test. Cosine between class means, hardest class first. Orange: similar. Blue: opposite.

- **Fixed basis rows: nearly orthogonal for every pair, mean |cos| 0.007.**
- **Learned outputs are not: mean |cos| 0.32 on VGG8, 0.68 on MobileNetV1.**
- **Animals group against vehicles; cat–dog is the closest pair on both.**

CONCLUSION

< 1%	12%	68%
extra MACs, nothing trained	of MACs: MobileNetV1	of MACs: VGG8

- A single head's margin is usually enough.
- The best training partitions also give the best exits.
- BP joint exits are better: LLS's case is training memory.
- Interpretable: calibrated heads, and they show where to merge.

Can local gradients scale beyond shallow CNNs?

$$M_{\text{BP}} \approx \sum_k M_k, \qquad M_{\text{local}} \approx \max_k M_k$$

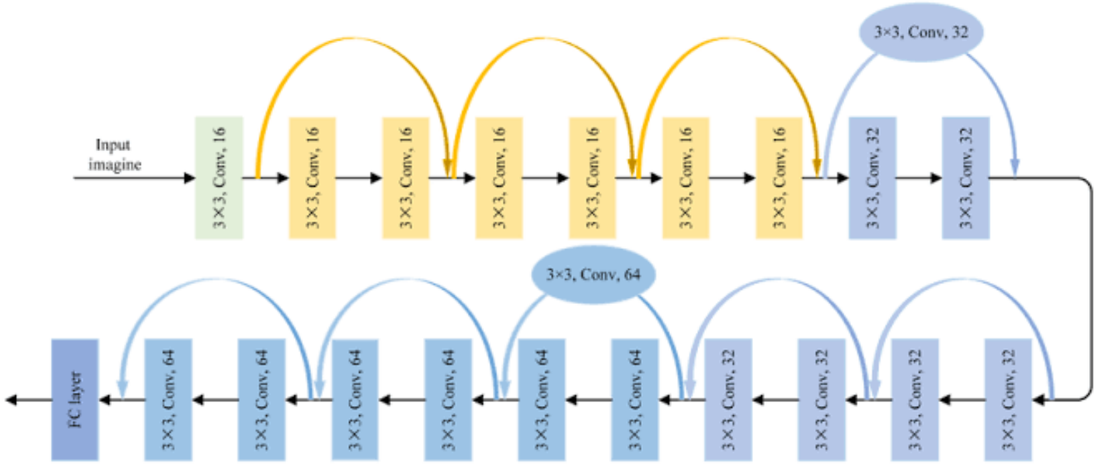
Residual networks

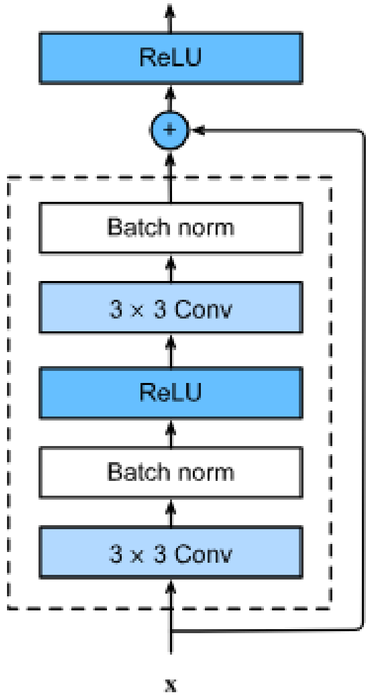
Can local loss preserve the skip path?

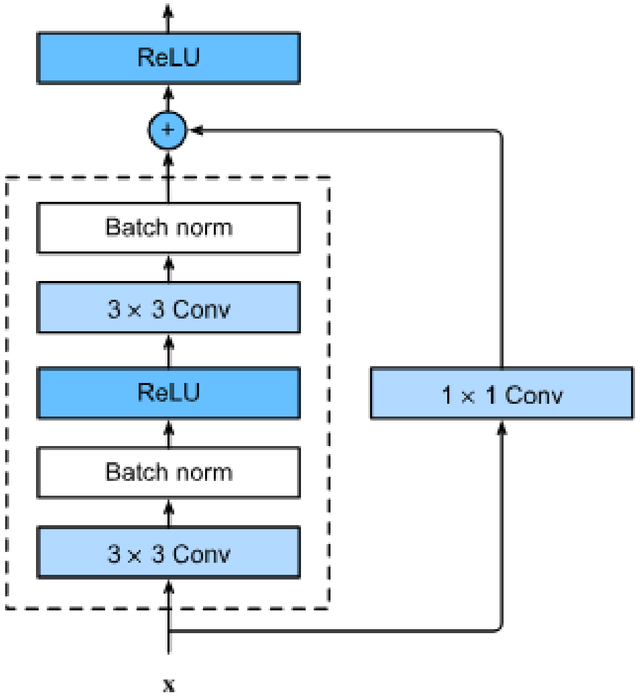
Vision Transformers

Can it supervise attention + tokens?

SCALABILITY: RESNET

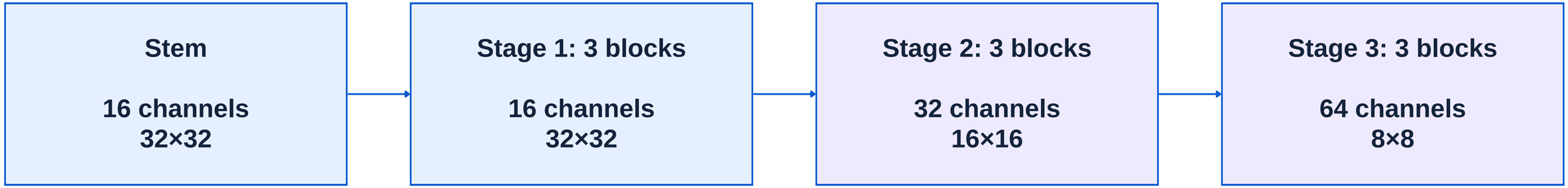






$$h = \phi(F(x; W) + S(x)), \qquad S(x) = x \text{ or } W_s x$$

ResNet-20 becomes ten training units

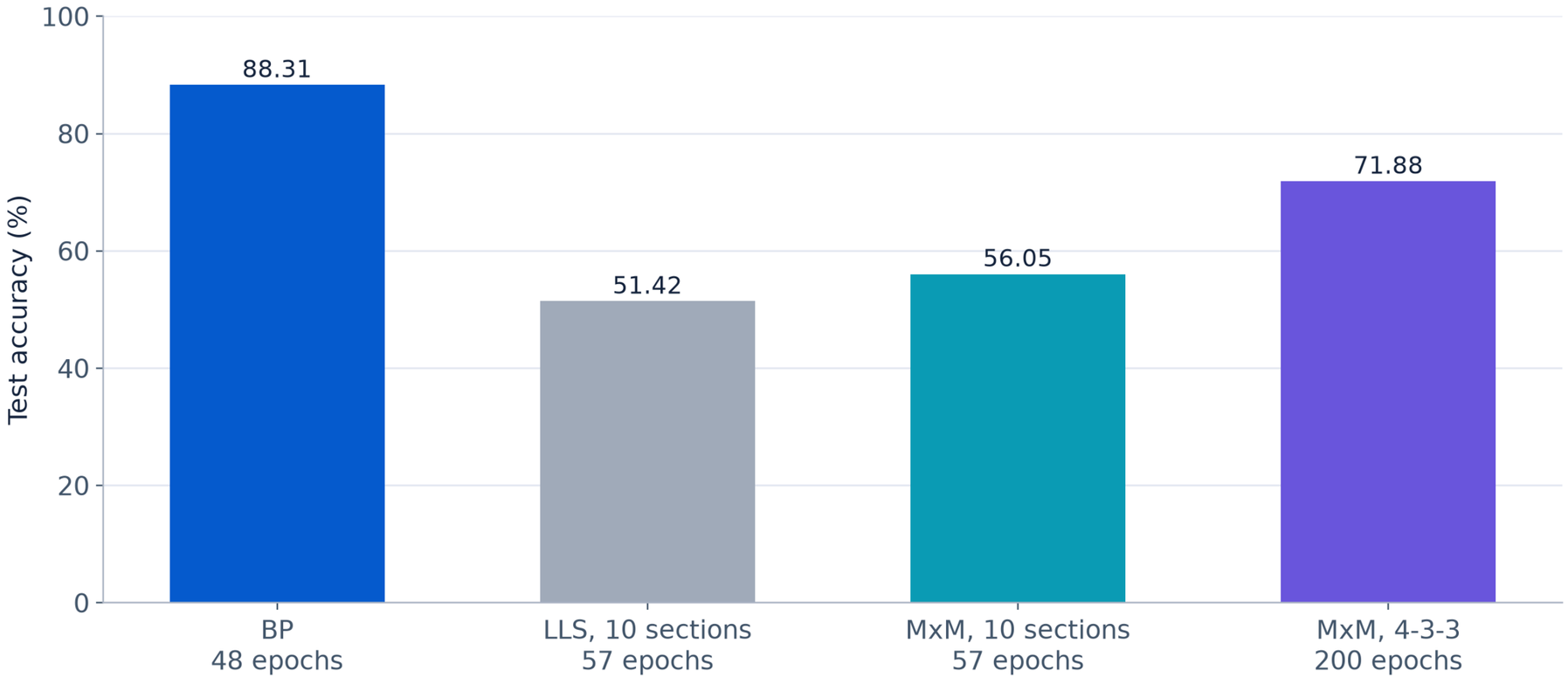


Partition	Local sections
1 × 10	stem + nine independent residual units
4-3-3	stem ; stage 2; stage 3
7-3	stem + stage 1+ stage 2; stage 3

The ResNet experiments we ran

Model / method	Partition	Readout	Epochs done
ResNet-20 BP	end-to-end	linear	48 / 100 (early stopping, patience=10)
ResNet-20 LLS	1 × 10	detached linear	57 / 100 (early stopping, patience=10)
ResNet-20 LLS-MxM	1 × 10	detached linear	57 / 100 (early stopping, patience=10)
ResNet-20 LLS-MxM	4-3-3	detached linear	200
Wide-ResNet-20 BP	end-to-end	linear	300
Wide-ResNet-20 LLS-MxM	7-3	last LLS head	300

ResNet-20: stronger LLS models still trail BP



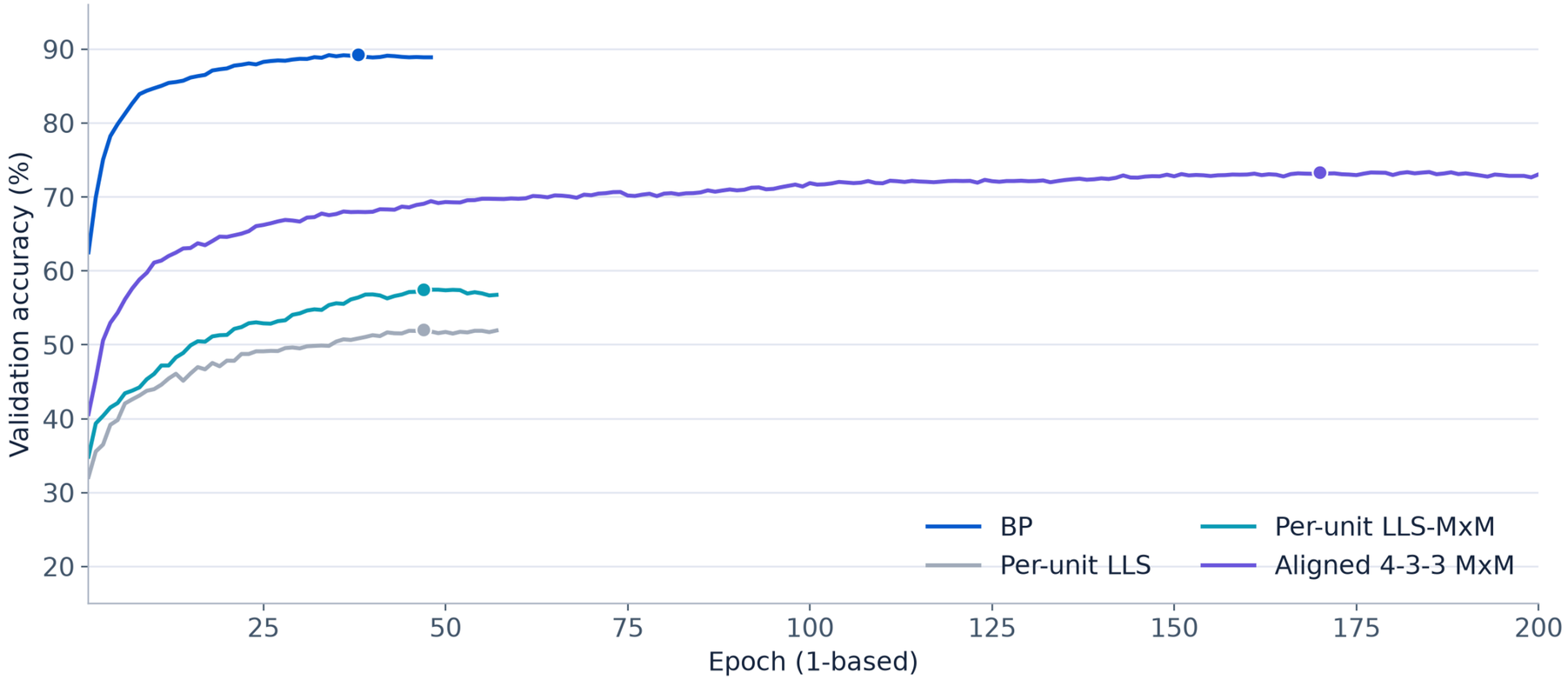
+4.63%

class mixing: MxM versus fixed per-unit LLS

+15.83%

aligned versus per-unit MxM; more training too

ResNet-20 learning trajectories

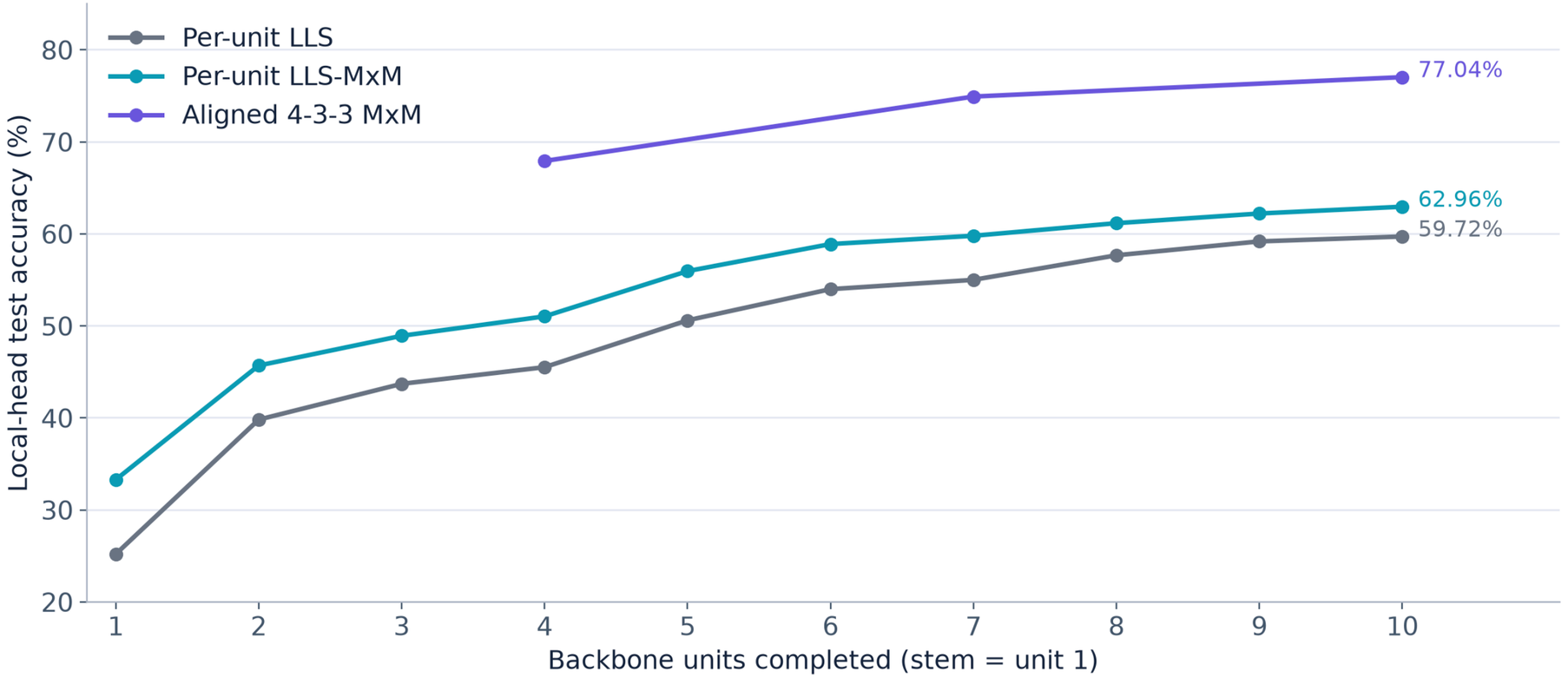


Early stopping
BP: 48 epochs
Per-unit: 57

Aligned training continues to 200 epochs.

Different budgets limit causal claims.

Where does each ResNet local head improve?



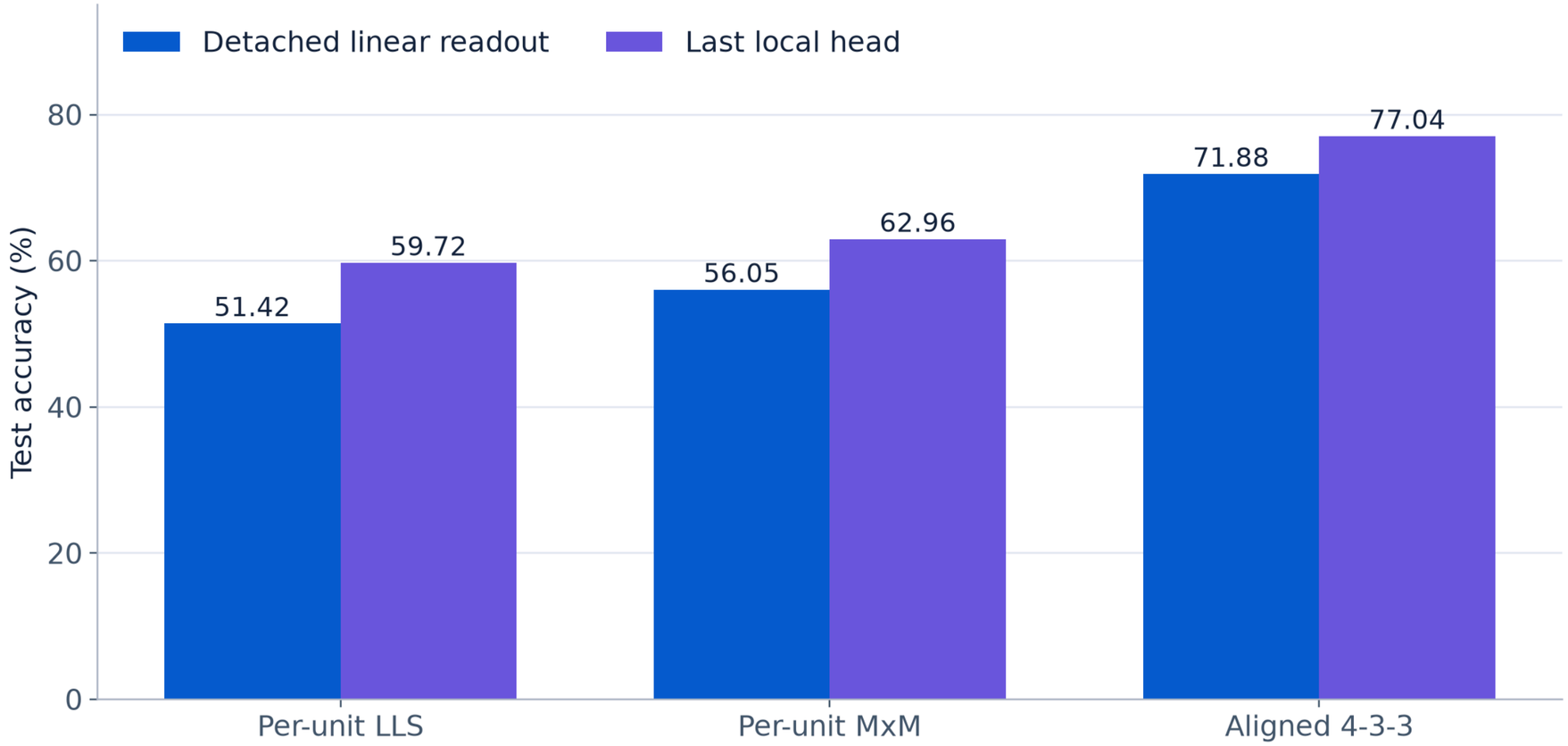
67.93%

first aligned head, after unit 4

51.04%

per-unit MxM head at the same depth

The final local head can beat the linear readout



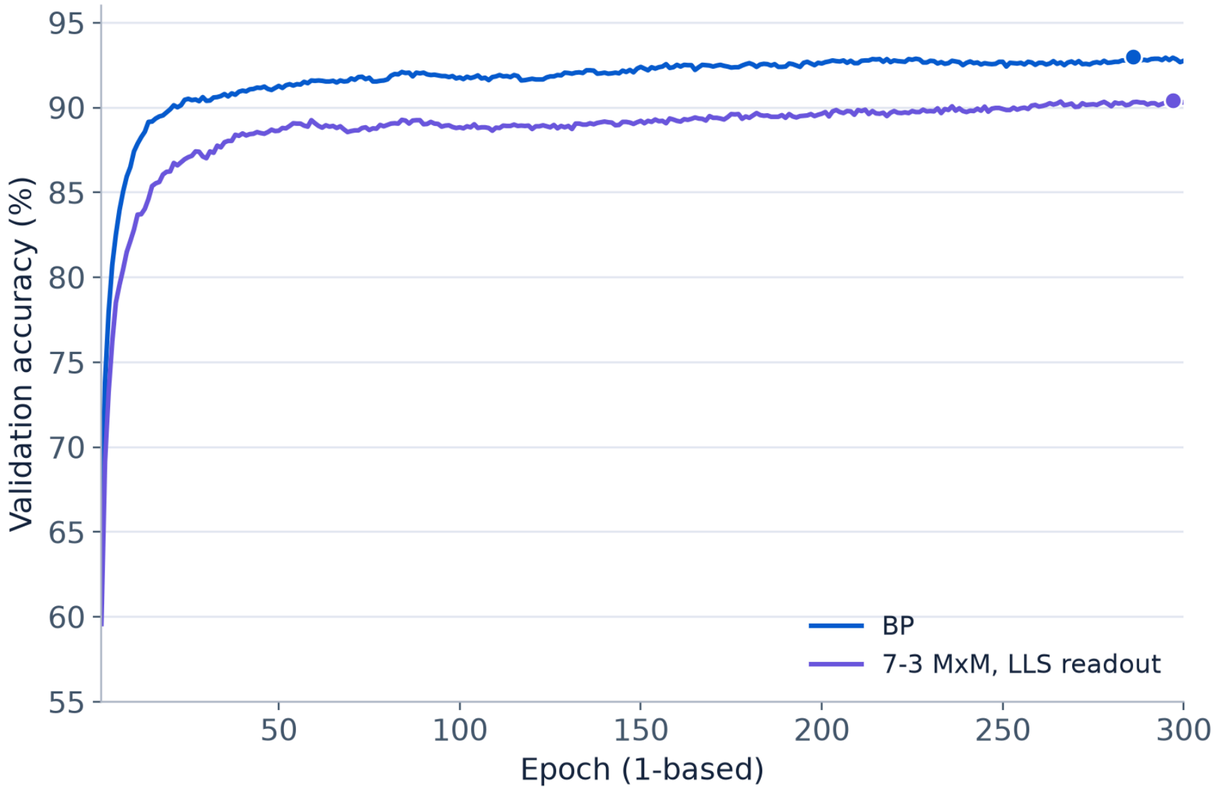
77.04%

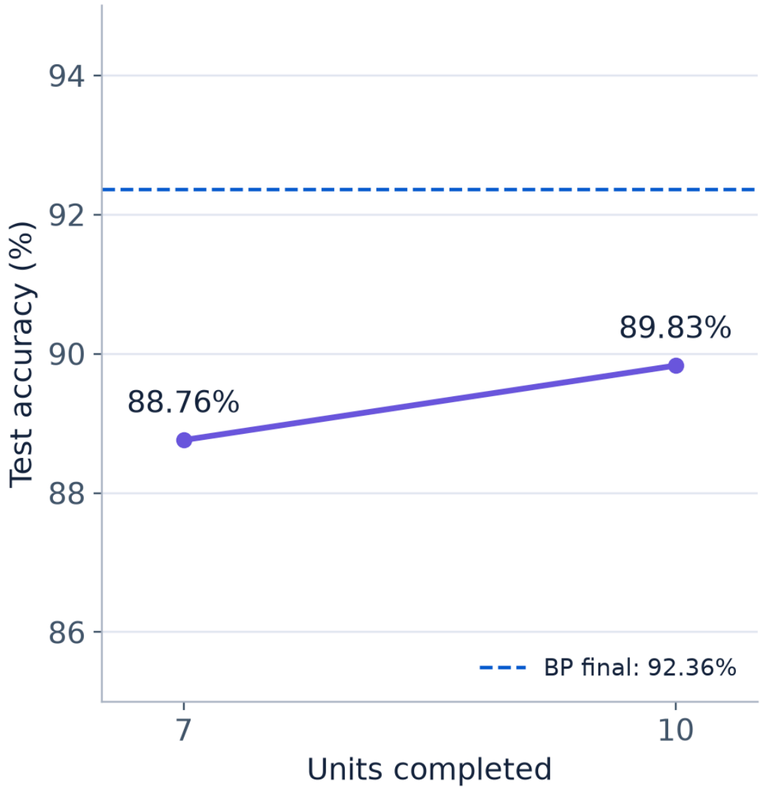
aligned model’s last local head

71.88%

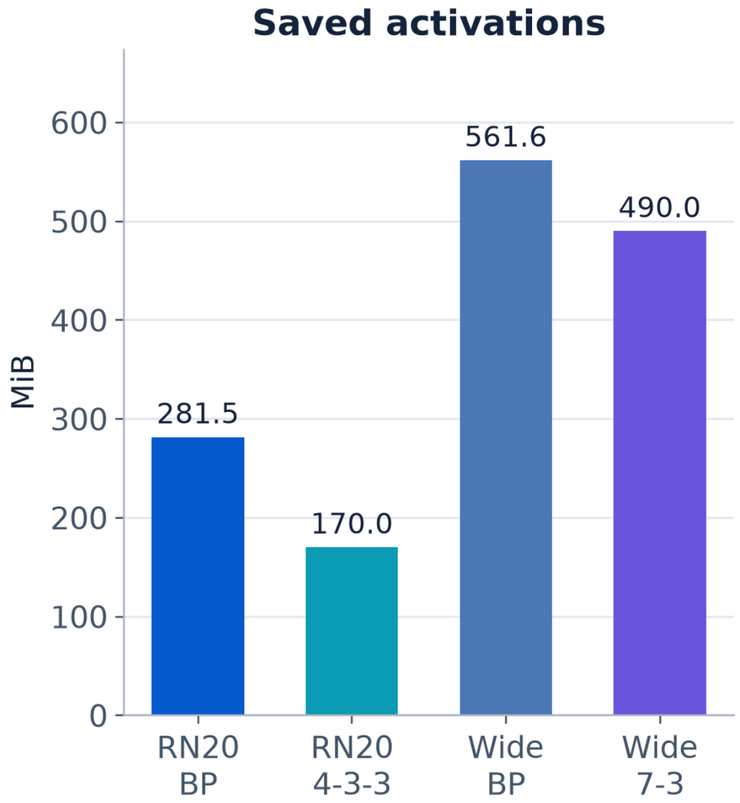
aligned model’s detached linear readout

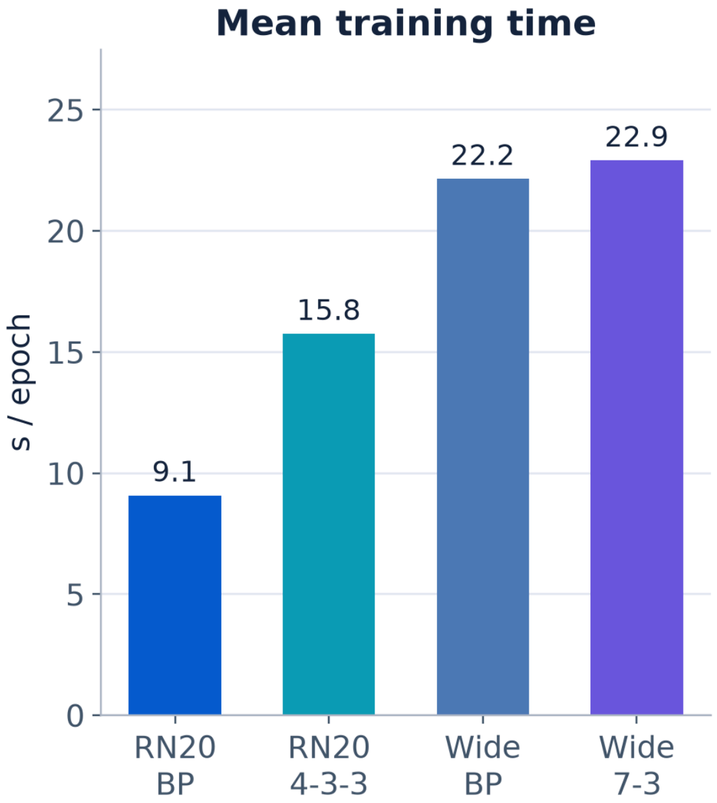
Wide resnet model

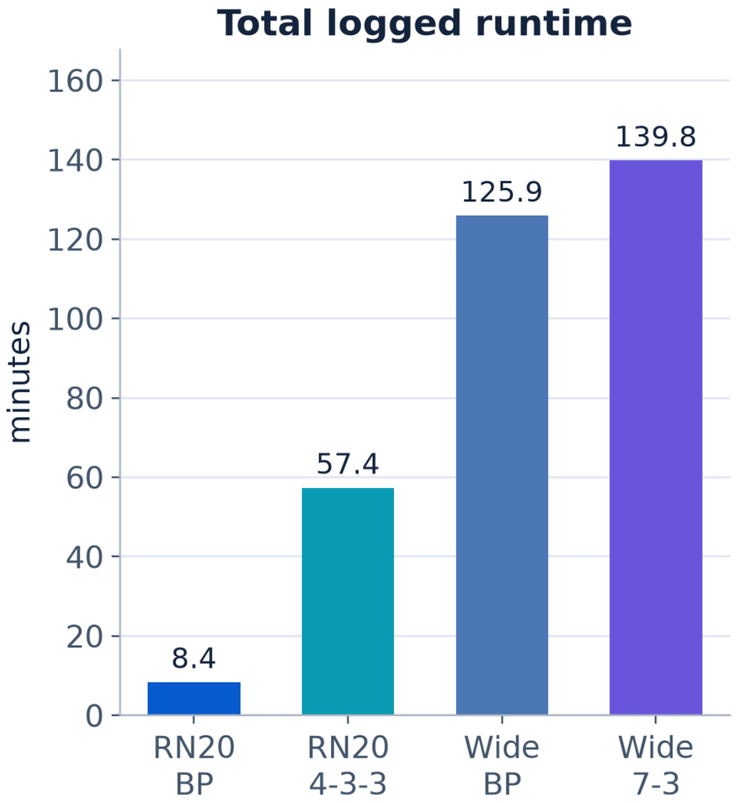




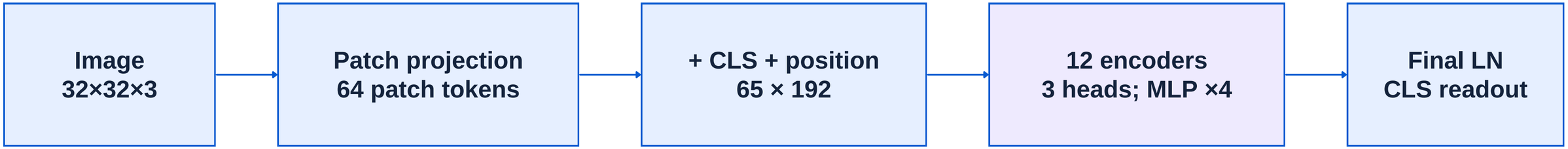
CS6886: Project Presentation





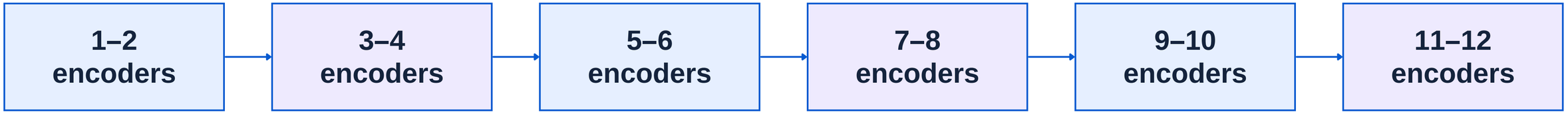


ViT-Tiny: the next test is self-attention



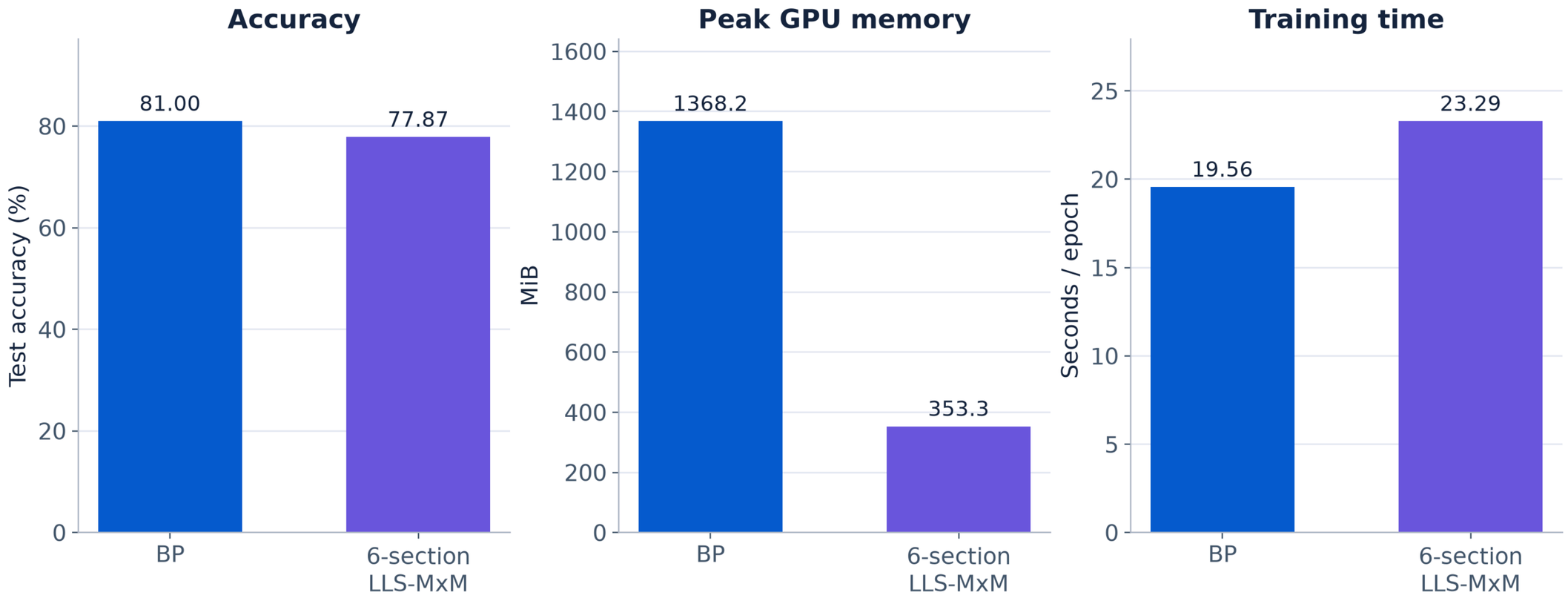
$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^{\mathsf{T}}}{\sqrt{d_k}}\right)V$$

A complete encoder is one ViT unit

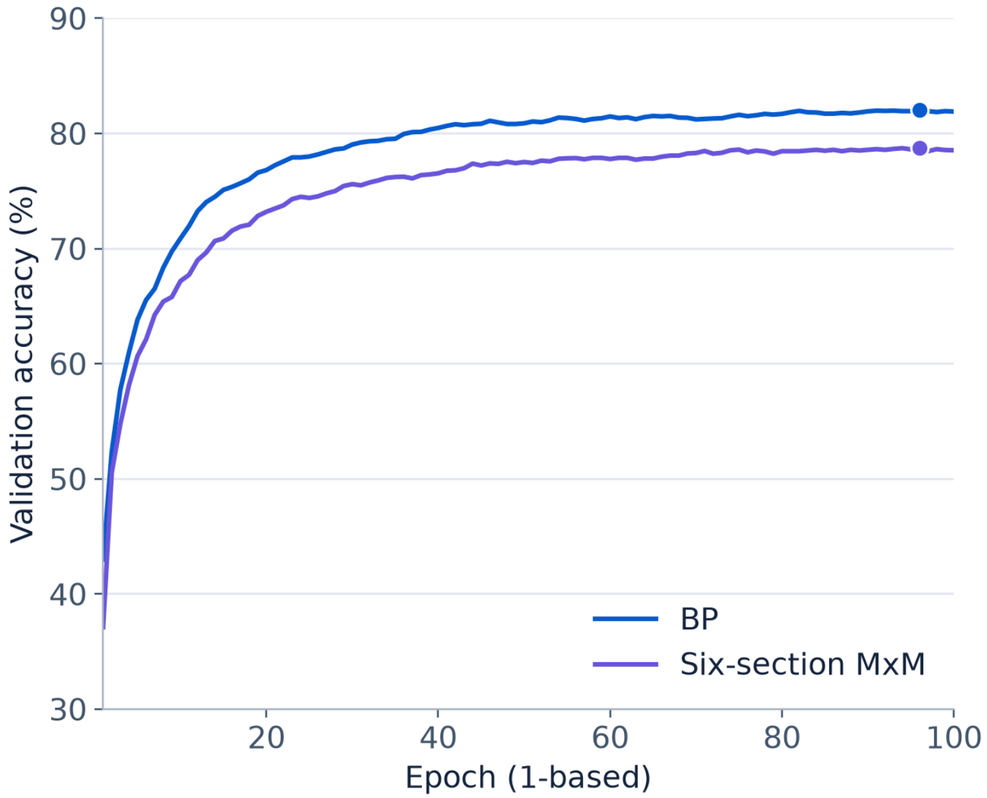


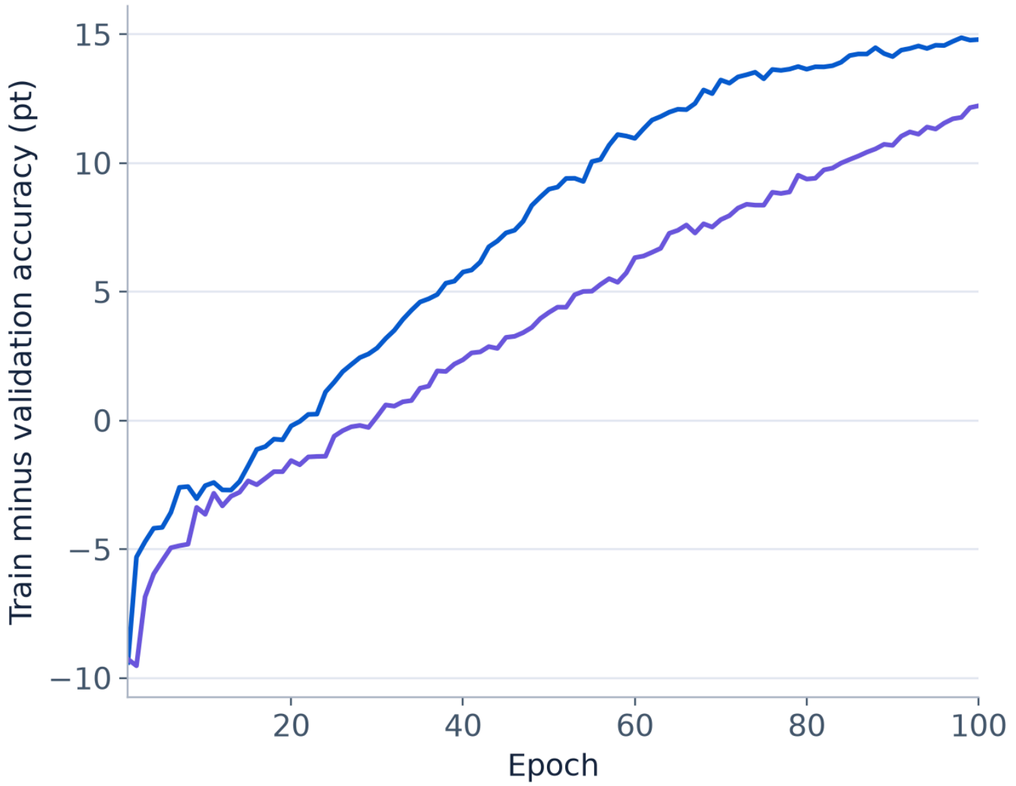
$$u = x + \mathrm{MSA}(\mathrm{LN}(x)), \qquad h = u + \mathrm{MLP}(\mathrm{LN}(u))$$

ViT: 3.13-point accuracy cost, 74.2% memory saving

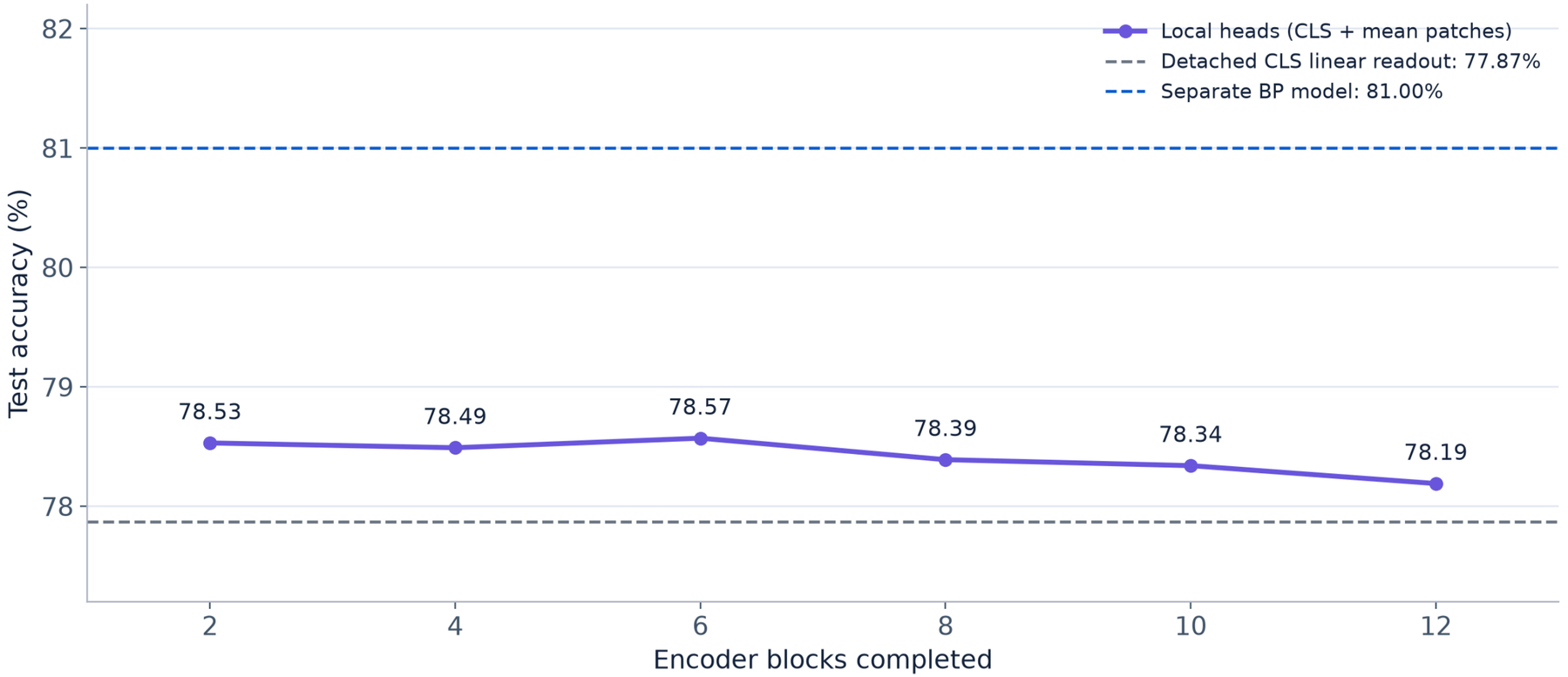


ViT learning curves and generalization gap





ViT heads plateau after the first section



78.53%

first local head after 2 encoders

78.19%

last local head after 12 encoders

Thank you